

Equivalent Fractions



We know that **fractions** can be compared when the **wholes are the same size**.

But can two different-looking fractions show the **same amount of one whole**?
Let's explore!

Let's Start

Here are **two**
wholes of the
same size.



Tap Next to divide both wholes into equal parts.

Next

To show $\frac{1}{2}$,
how many
equal parts
should
each whole be
divided into?



After selecting 2, animate both wholes dividing into 2 equal parts.

5

2

3

Tap the correct answer.

Next

Look at the
denominator of $\frac{1}{2}$.

It tells the
total number of
equal parts.



5

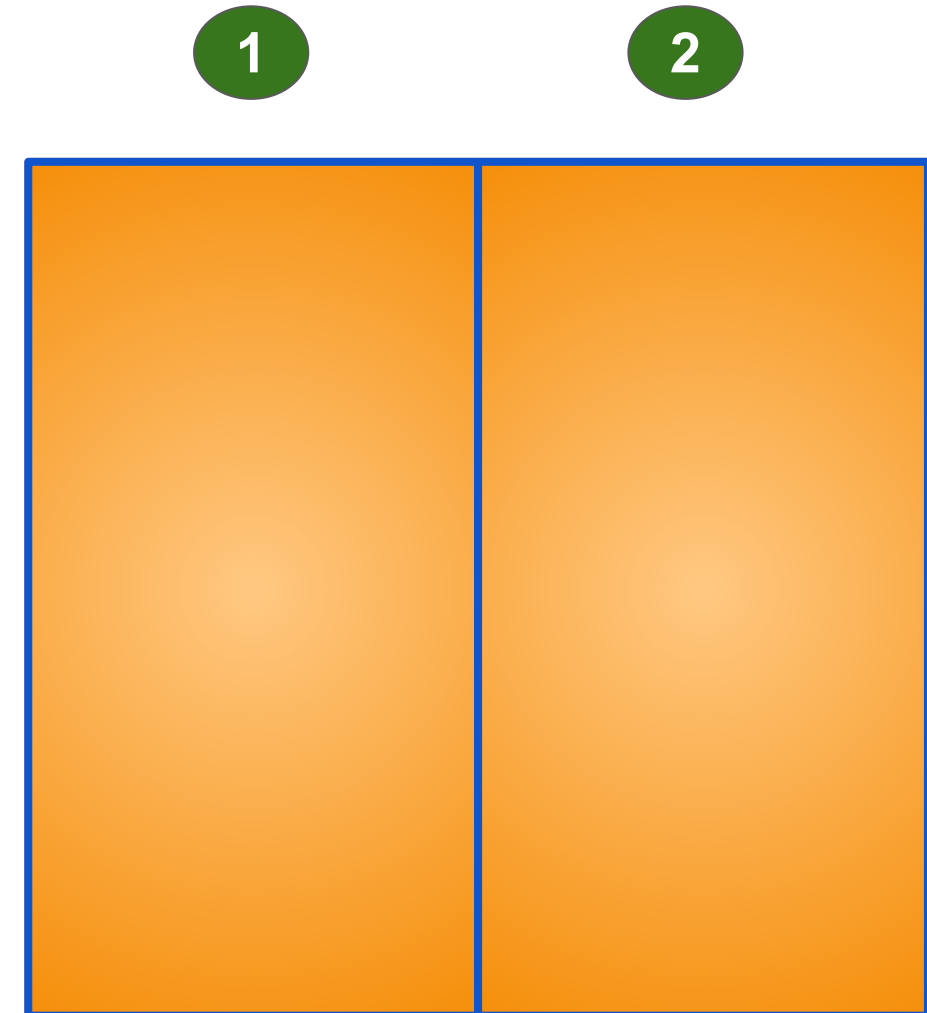
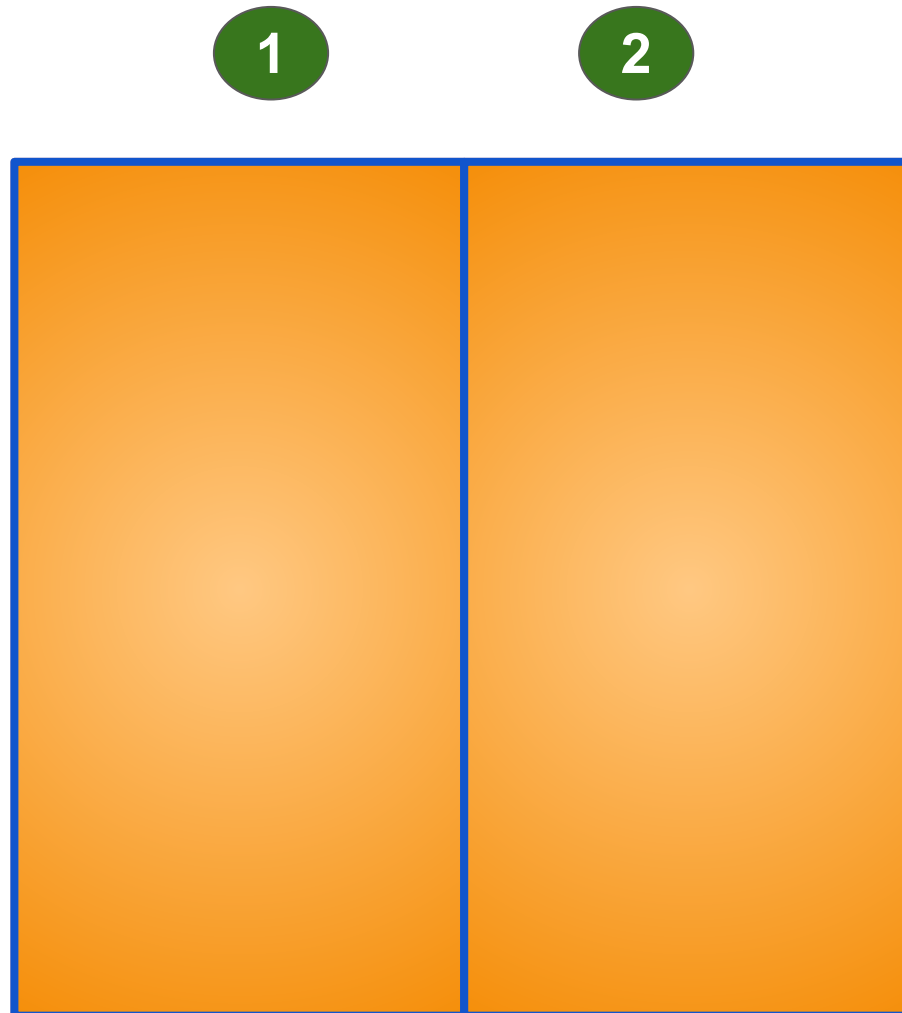
2

3

Tap the correct answer.

Next

Now,
shade $\frac{1}{2}$ of
each whole.

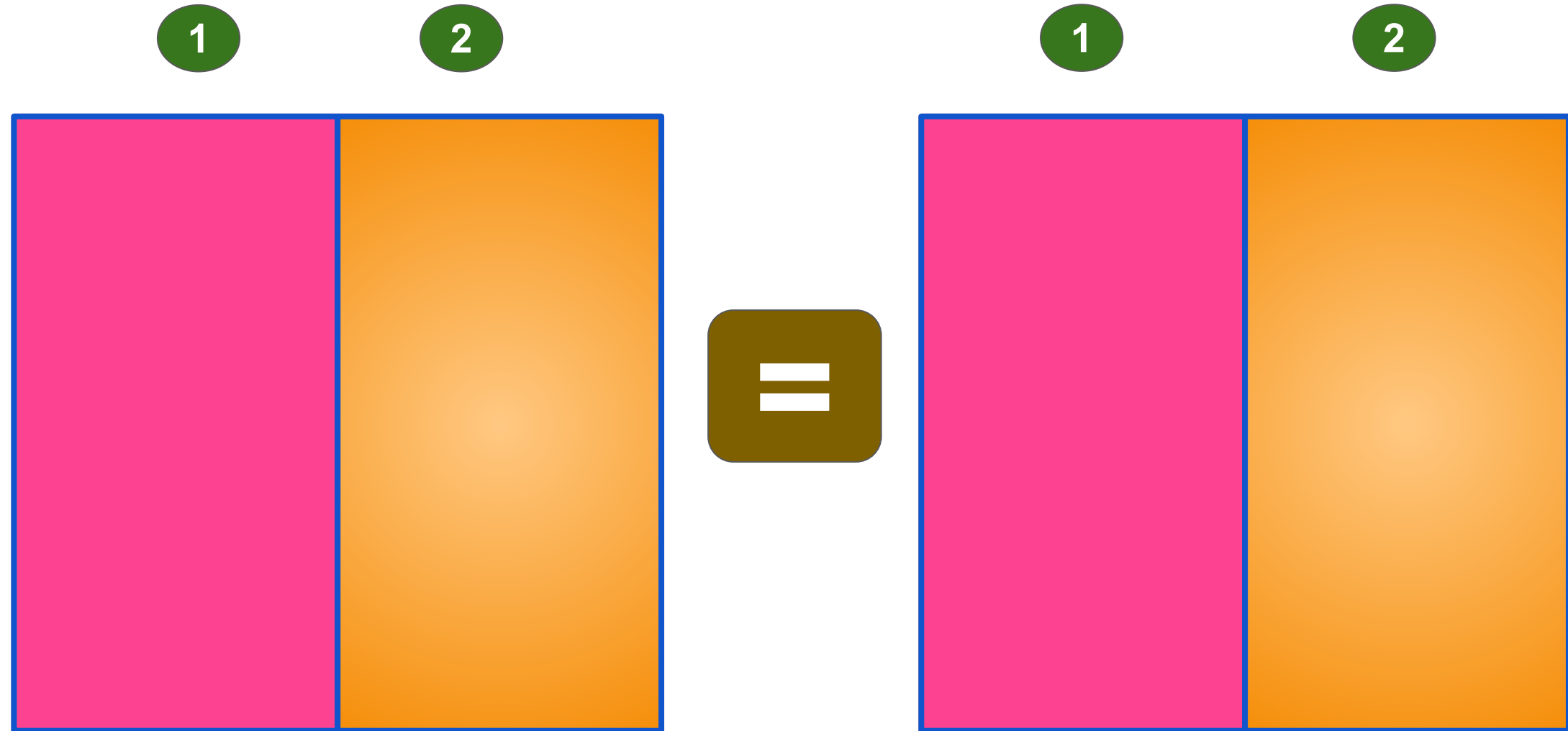


When the learner taps a part, briefly pulsate the selected part and fill it with the chosen colour. Require one shaded part in each 2-part whole before enabling Next.

Tap to choose the color. Then tap one part in each whole.

Next

So, both **wholes**
show
the fraction $\frac{1}{2}$.



$$\frac{1}{2}$$

Pulsate both pink parts first along
with equal sign and then pulsate both
 $\frac{1}{2}$ along with equal sign.

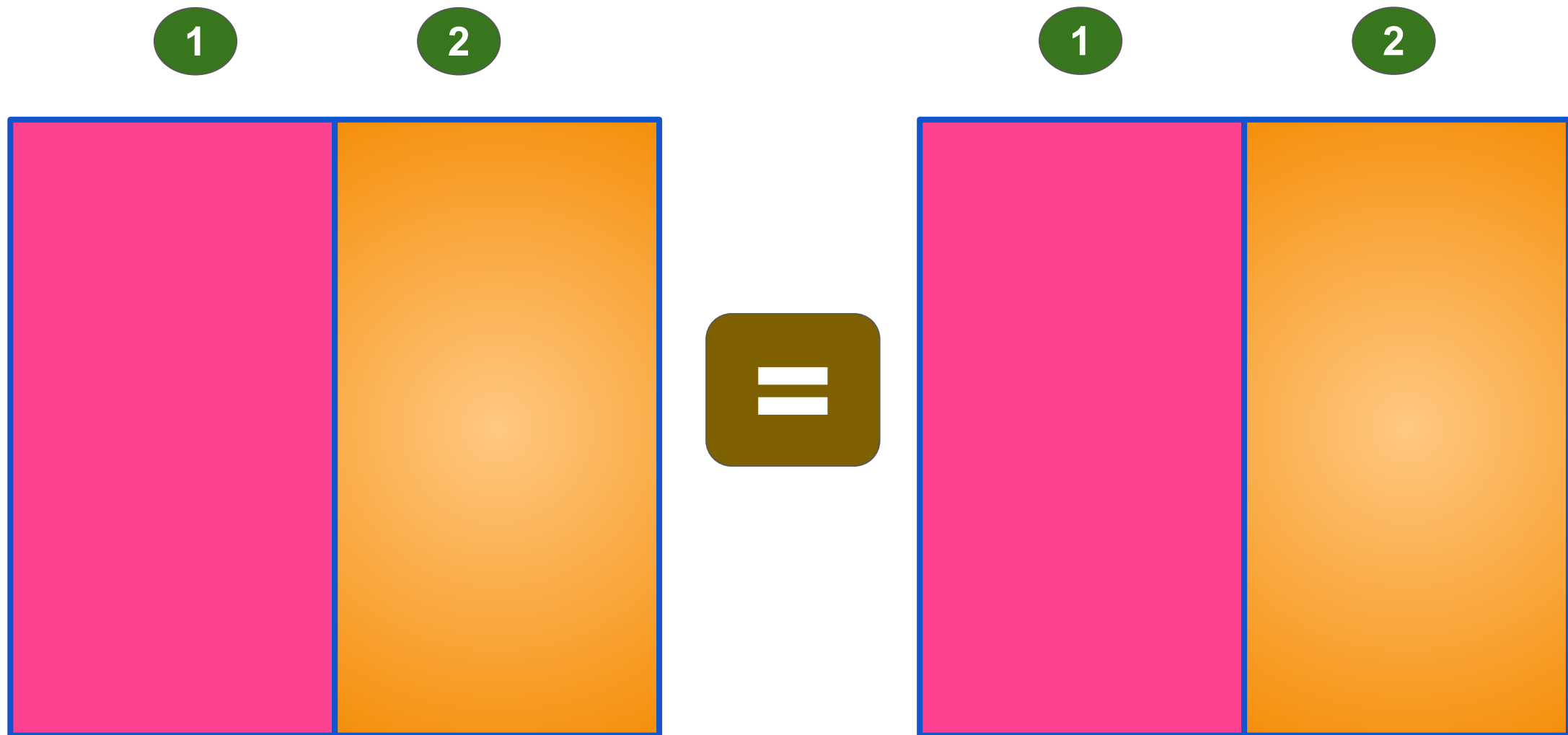
$$\frac{1}{2}$$

Tap next to continue.

Next

The **wholes** are the
same size and
divided into **equal**
parts.

The **same amount**
is **shaded in both**.



$$\frac{1}{2}$$

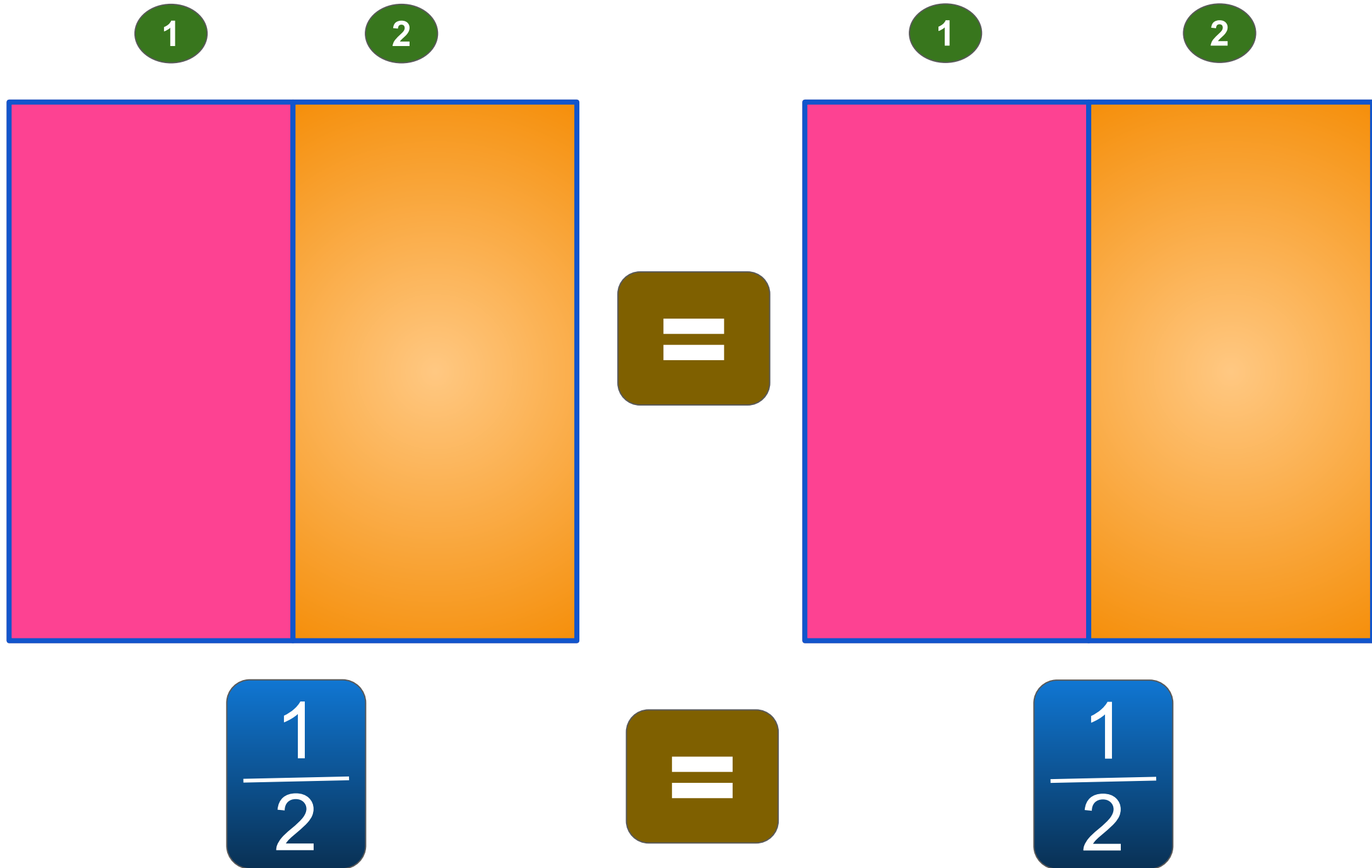
Pulsate both the wholes and then make
an animation of overlapping both the
wholes.

$$\frac{1}{2}$$

Tap Next

Next

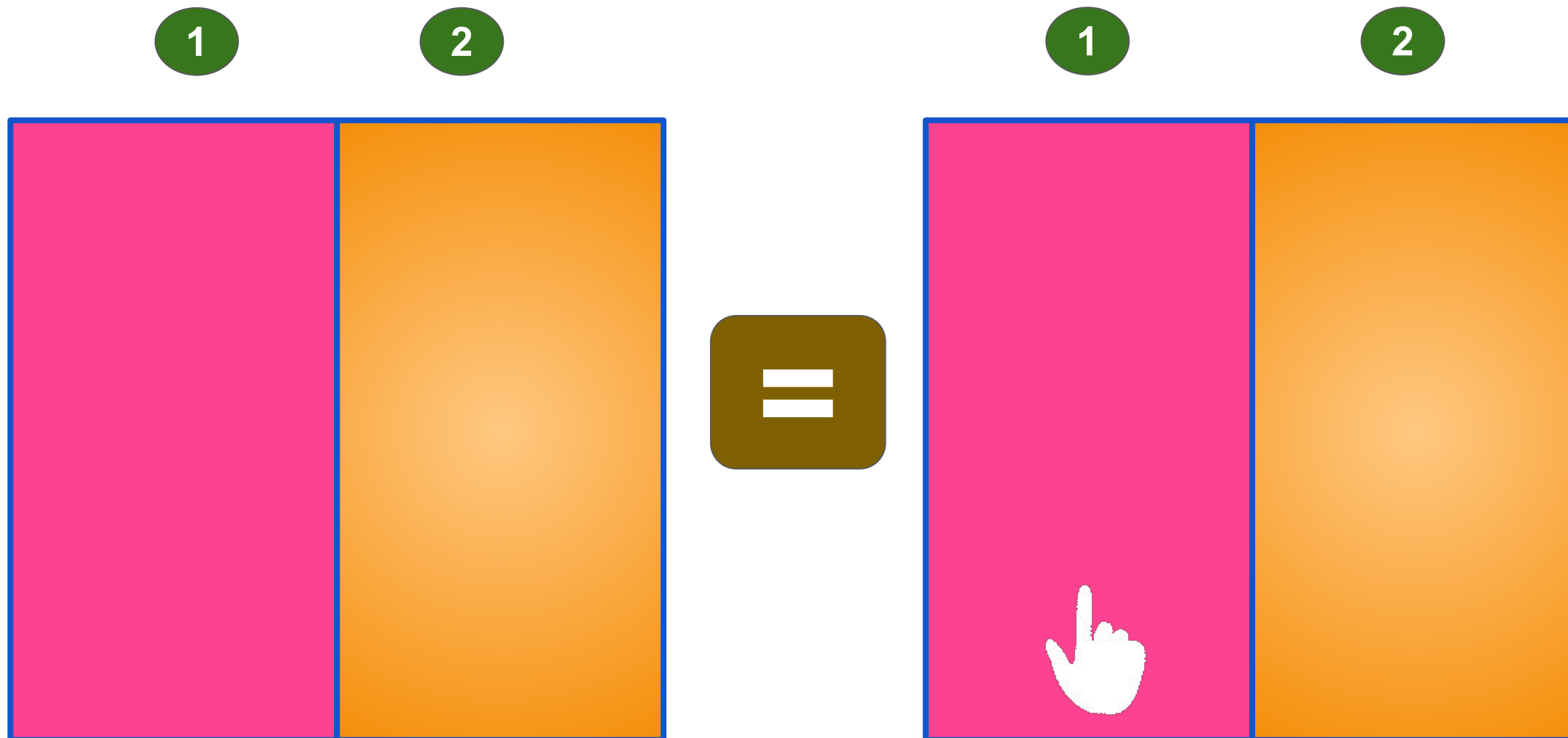
What happens if
we divide **one of**
the wholes into
smaller equal
parts?



Tap next to check.

Next

Let's divide one
whole into
smaller equal
parts.



$$\frac{1}{2}$$

Pulsate both wholes. Allow the learner to tap either of the whole. When the user chose one, dehighlight the unselected whole as like in the next slide

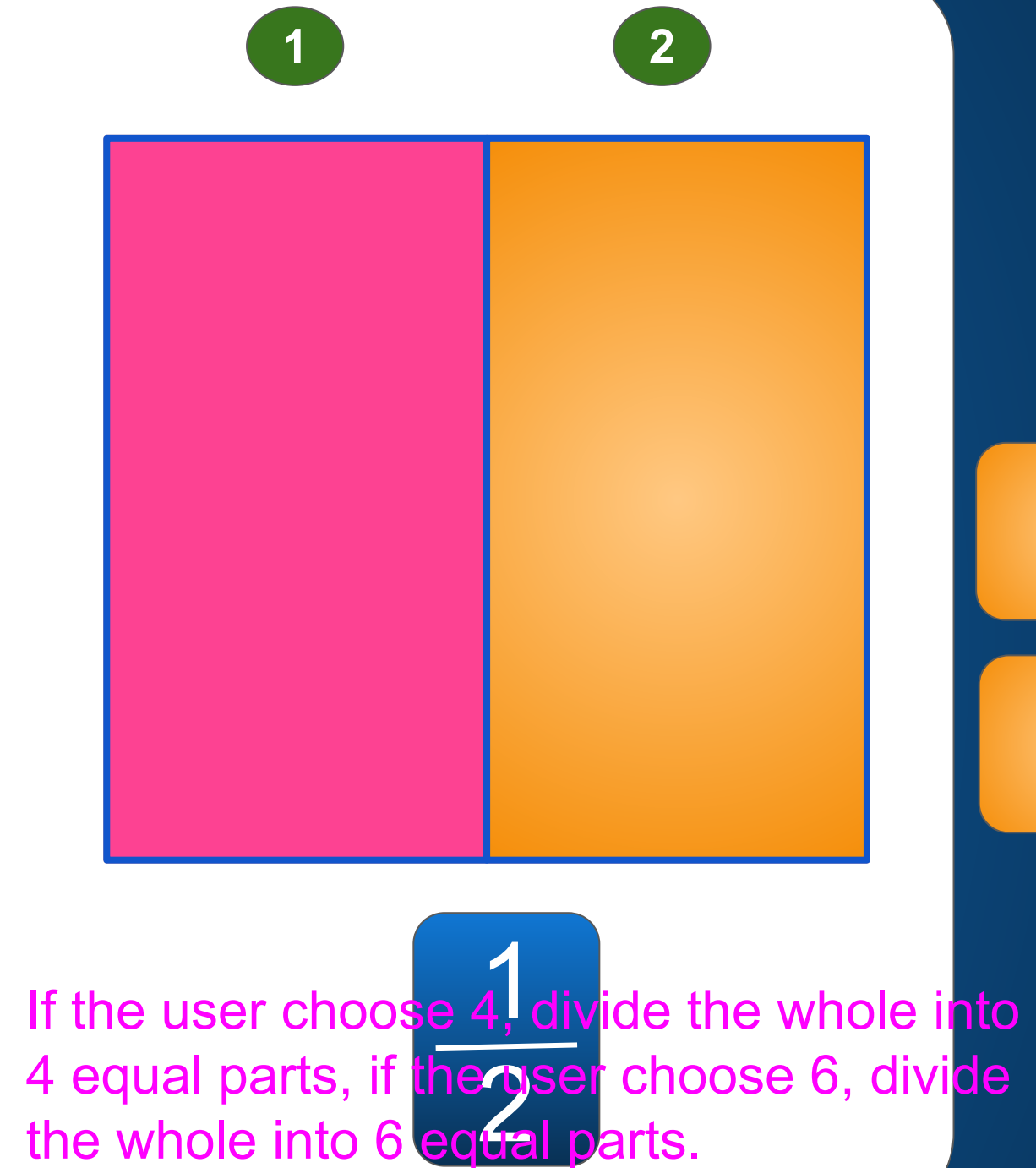
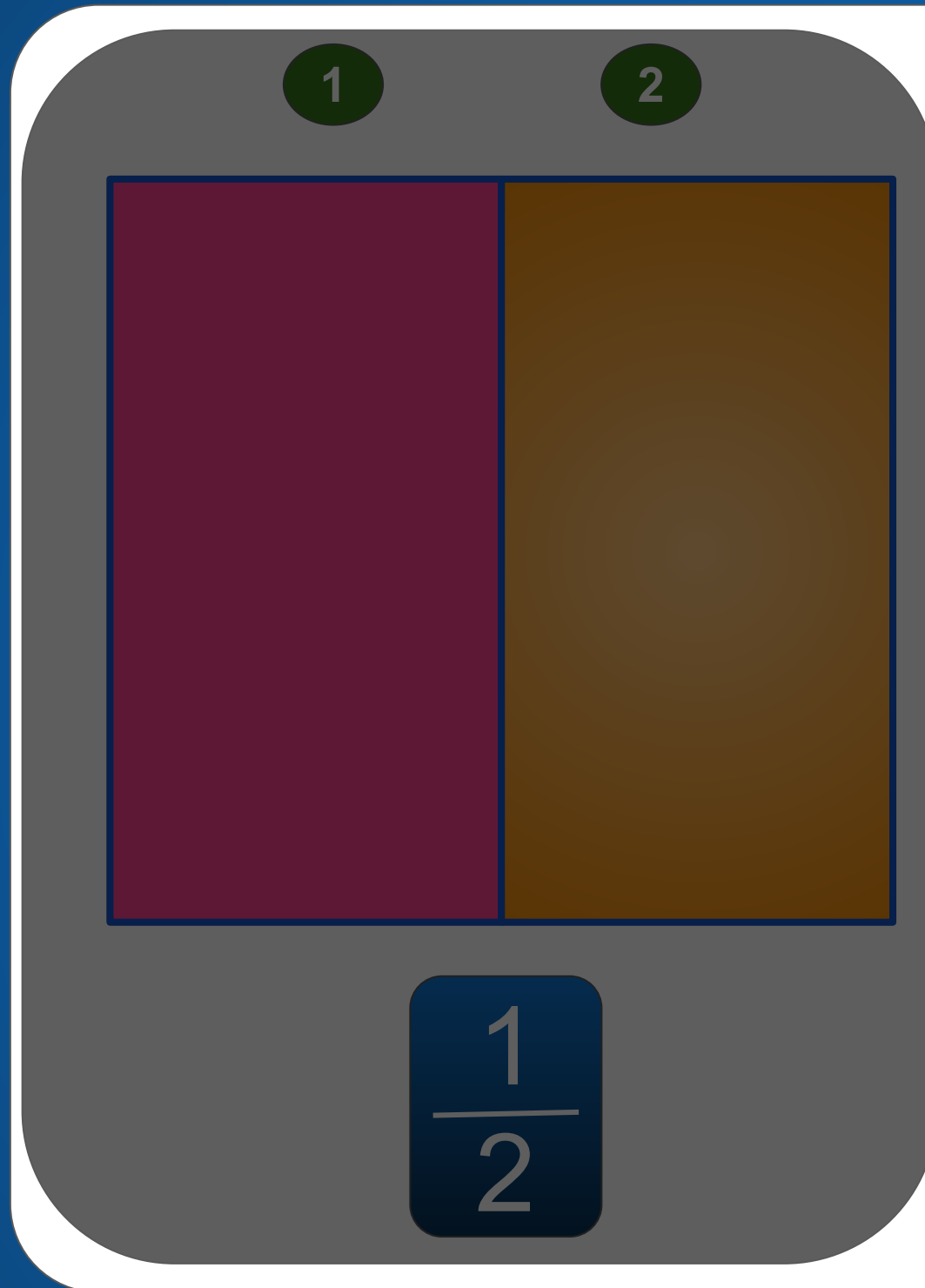


$$\frac{1}{2}$$

Tap either whole to choose it.

Next

Choose the number of **equal parts** to divide the **whole** into smaller equal parts.



4

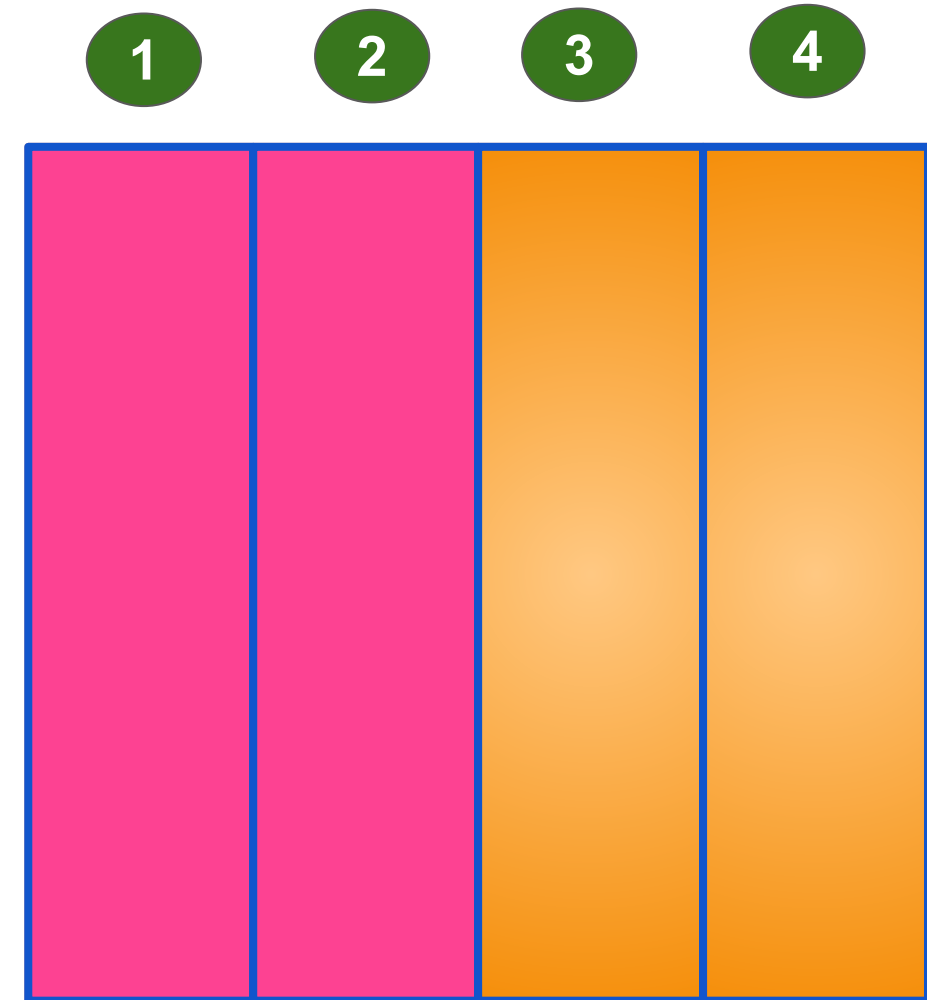
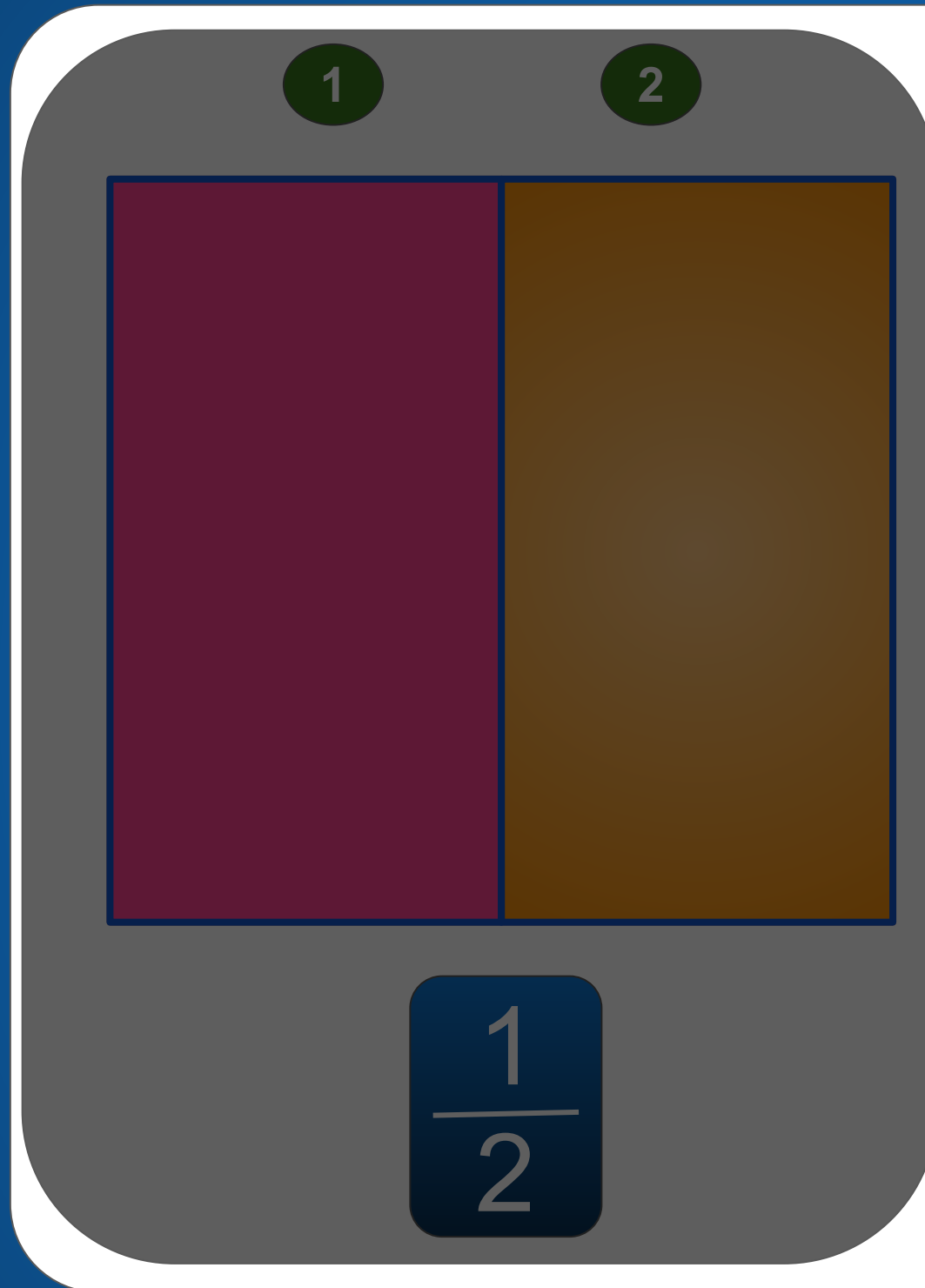
6

Tap 4 or 6 to divide.

Next

The whole has **4**
equal parts.

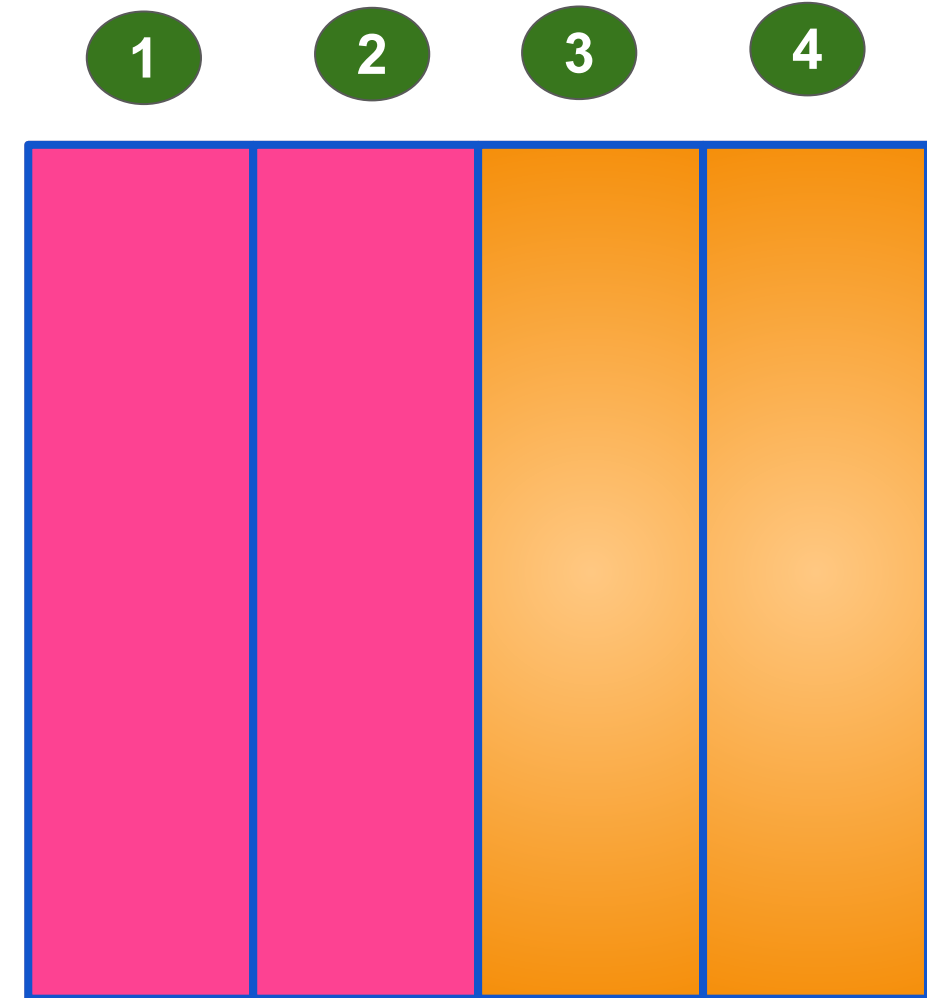
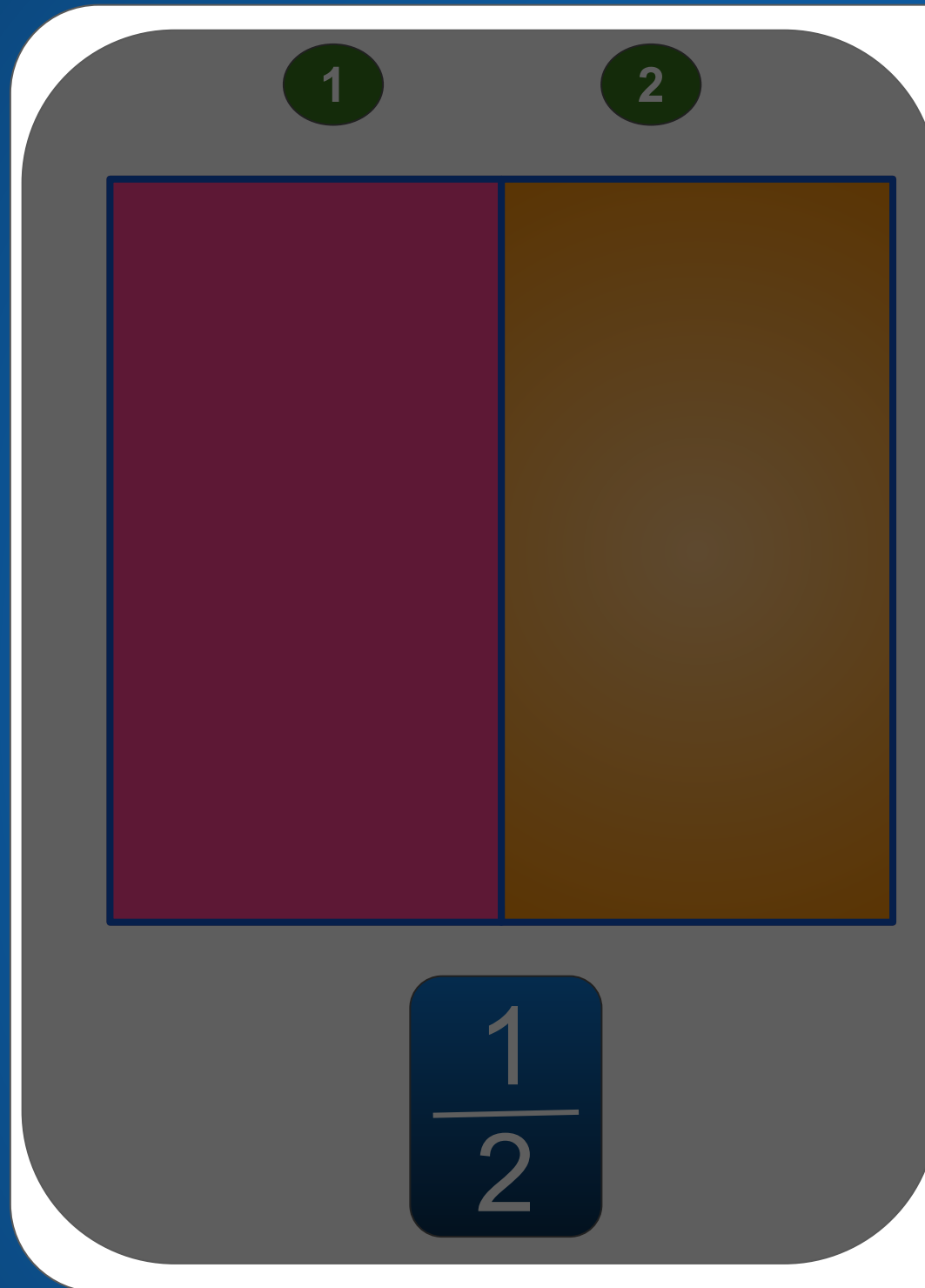
Each half is split
into
2 smaller parts.



Tap next to represent in fraction.

Next

Look at the divided **whole**.
What **fraction** is shaded now?



$$\frac{1}{4}$$

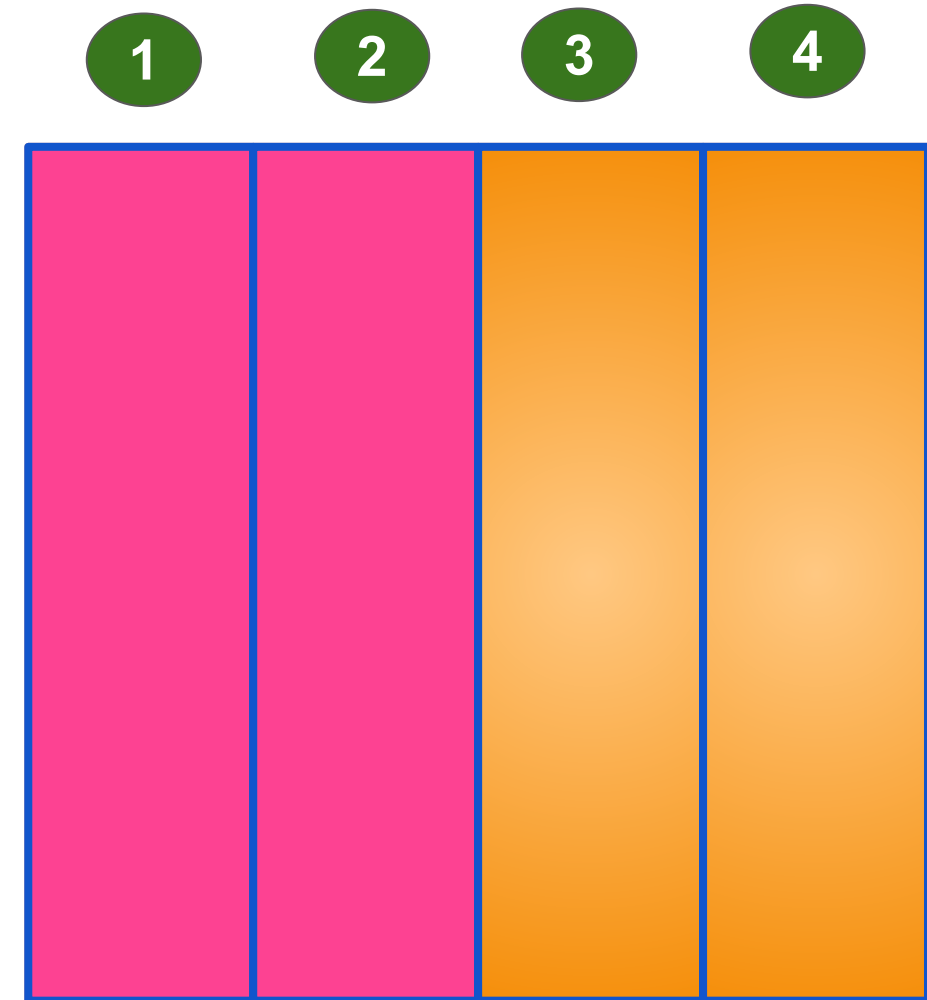
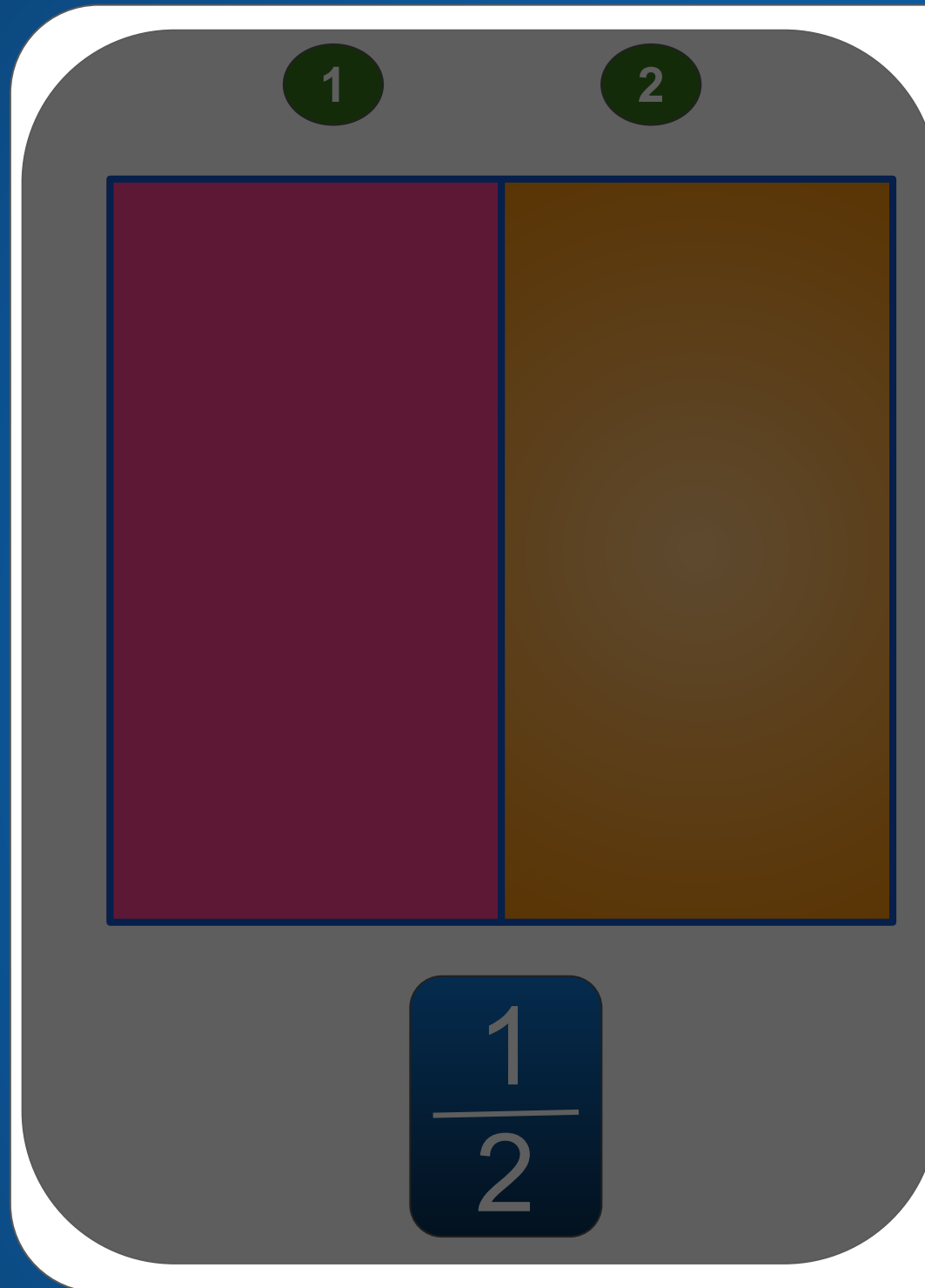
$$\frac{2}{4}$$

$$\frac{4}{2}$$

Tap the correct answer.

Next

Let's count again.
Check how many
parts are shaded?
And how many
equal parts are
there in all?



If the user chooses incorrect option, pulse 2 shaded parts allowing the user to count and then pulse all the 4 parts allowing user to count

$$\frac{1}{4}$$

$$\frac{2}{4}$$

$$\frac{4}{2}$$

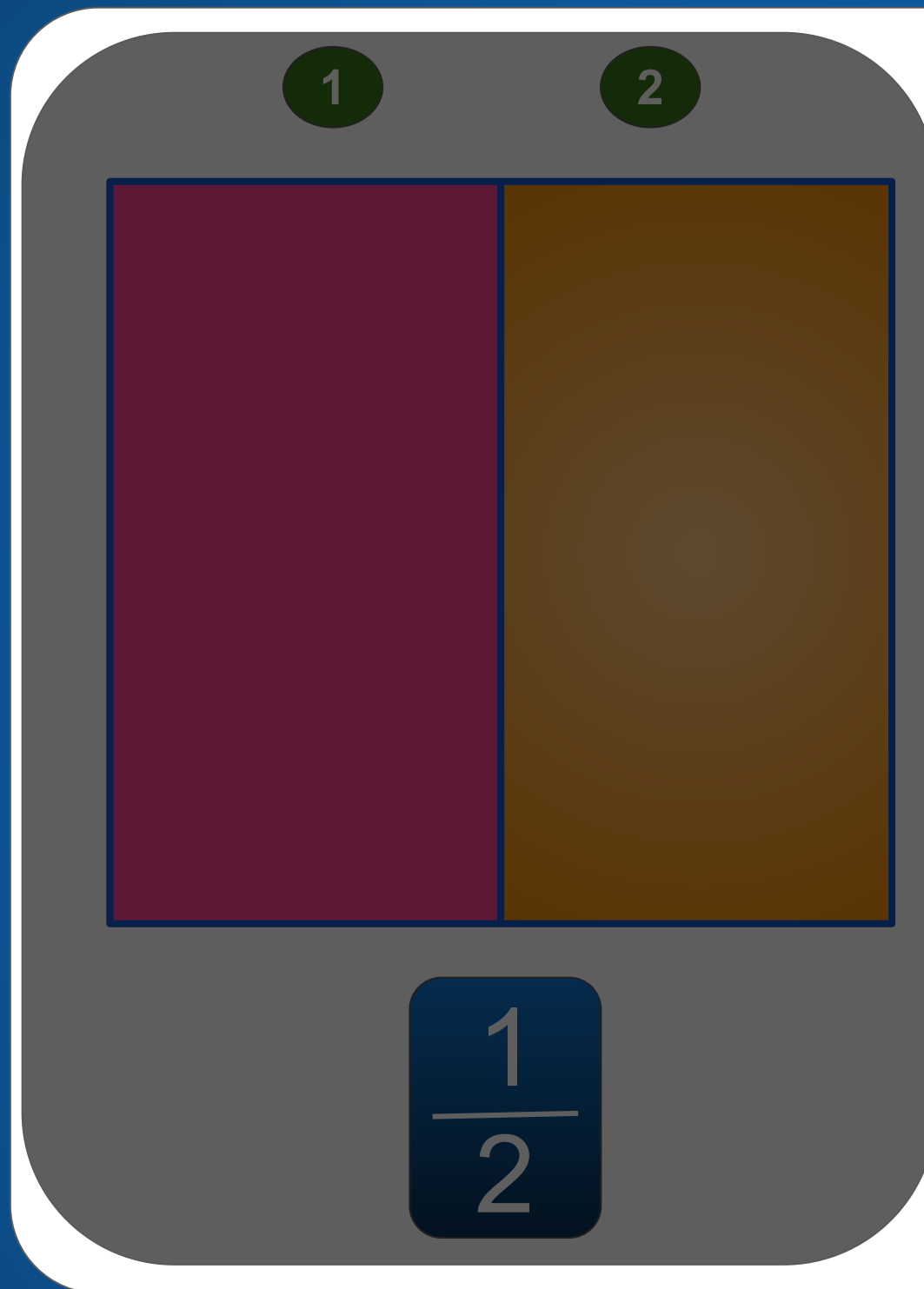
Tap the correct answer.

Next

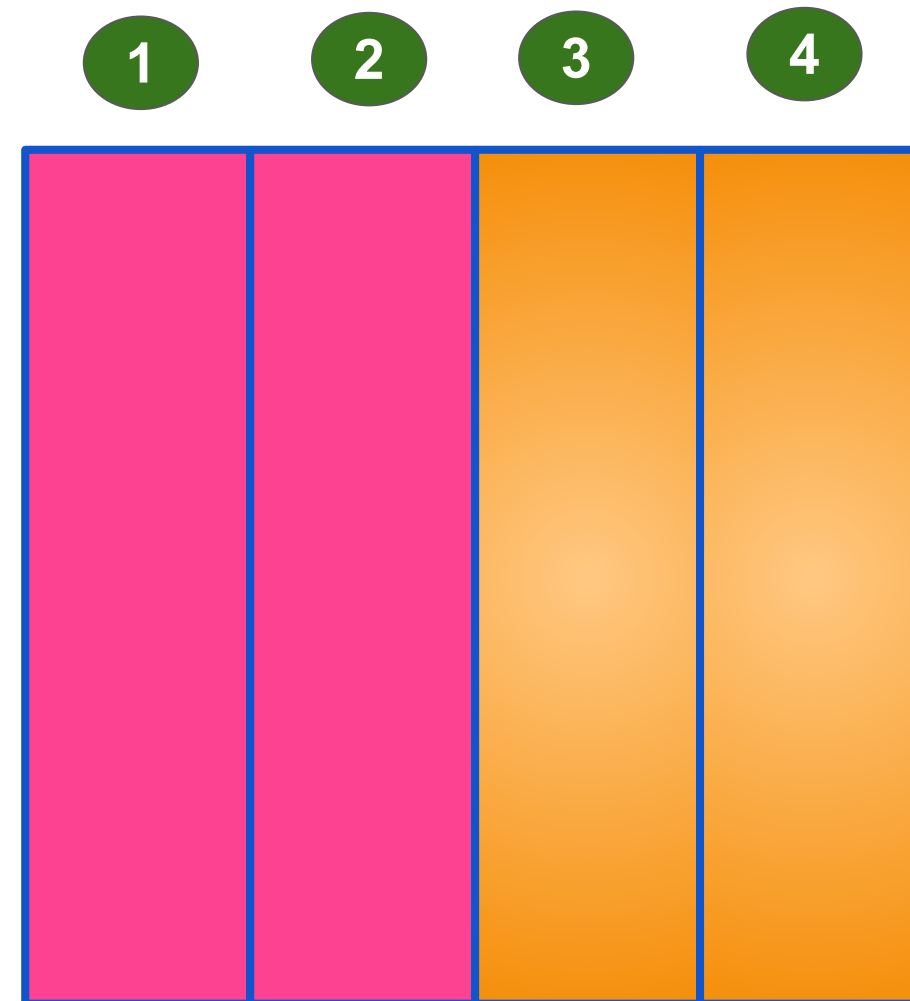
The **denominator 4** tells us the whole has **4 equal parts**.

The **numerator 2** tells us **2 parts** are shaded.

So, the **fraction** is $\frac{2}{4}$



If the user chooses correct option, pulse 2 shaded parts allowing the user to count and then pulse all the 4 parts allowing user to count



$$\frac{1}{4}$$

$$\frac{2}{4}$$

$$\frac{4}{2}$$

Tap next to continue.

Next

Now the **fractions**

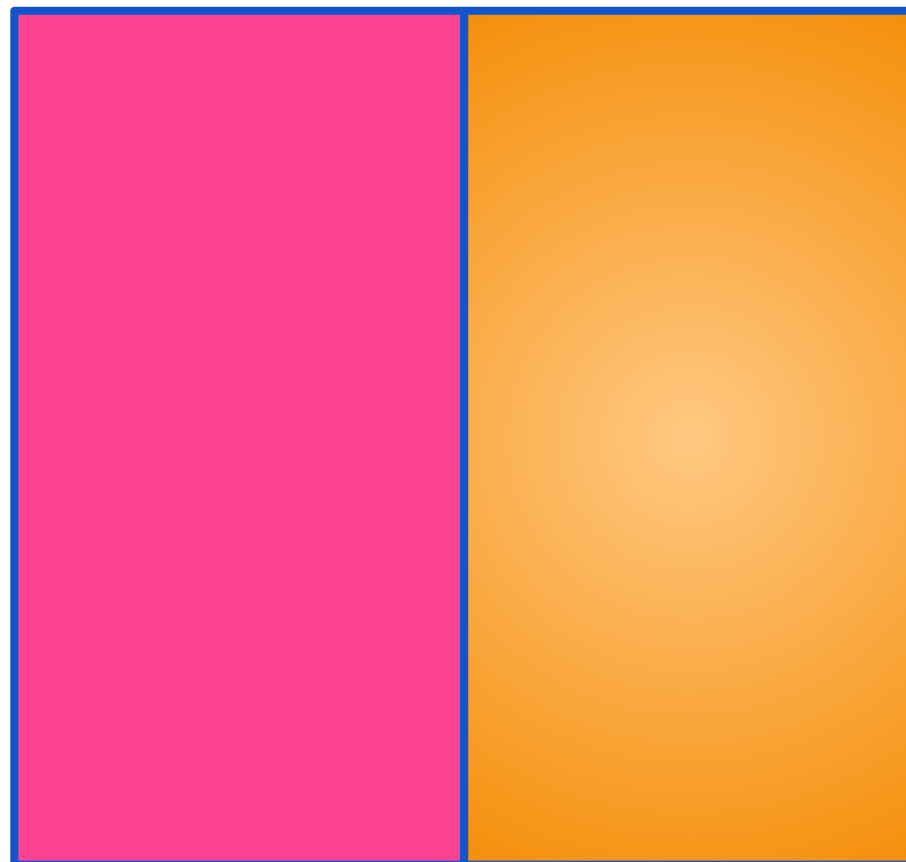
look different.

$\frac{2}{4}$ has **2 shaded parts**, but $\frac{1}{2}$ has only **1 shaded part**.
Is $\frac{2}{4}$ greater than $\frac{1}{2}$?



1

2



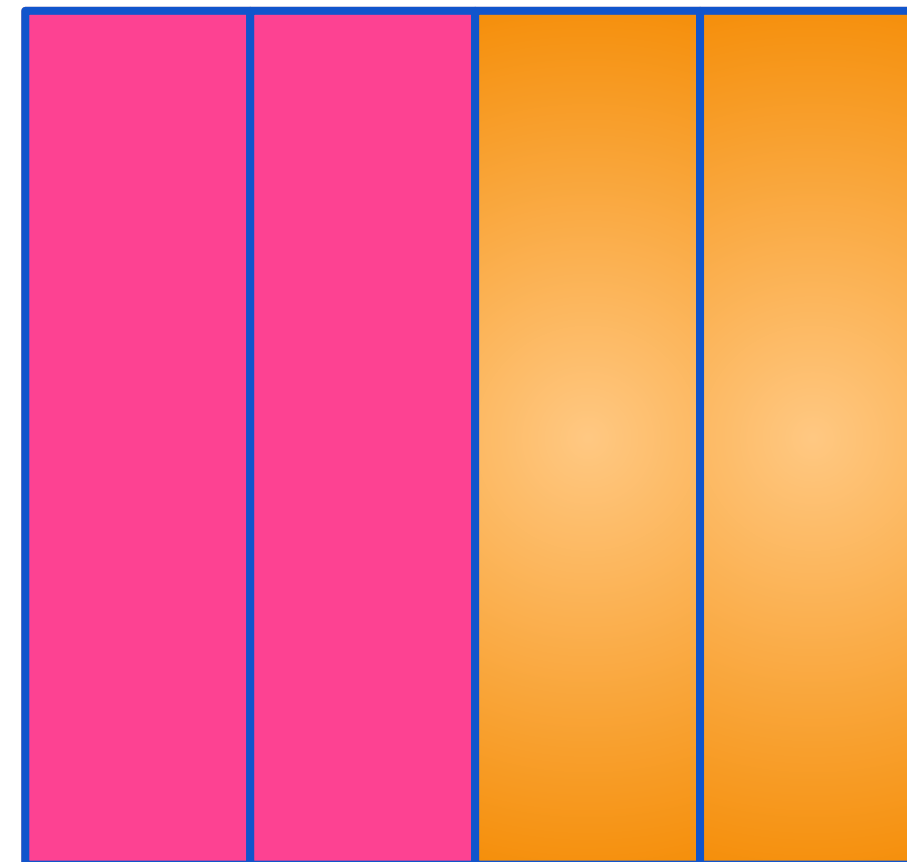
$$\frac{1}{2}$$

1

2

3

4

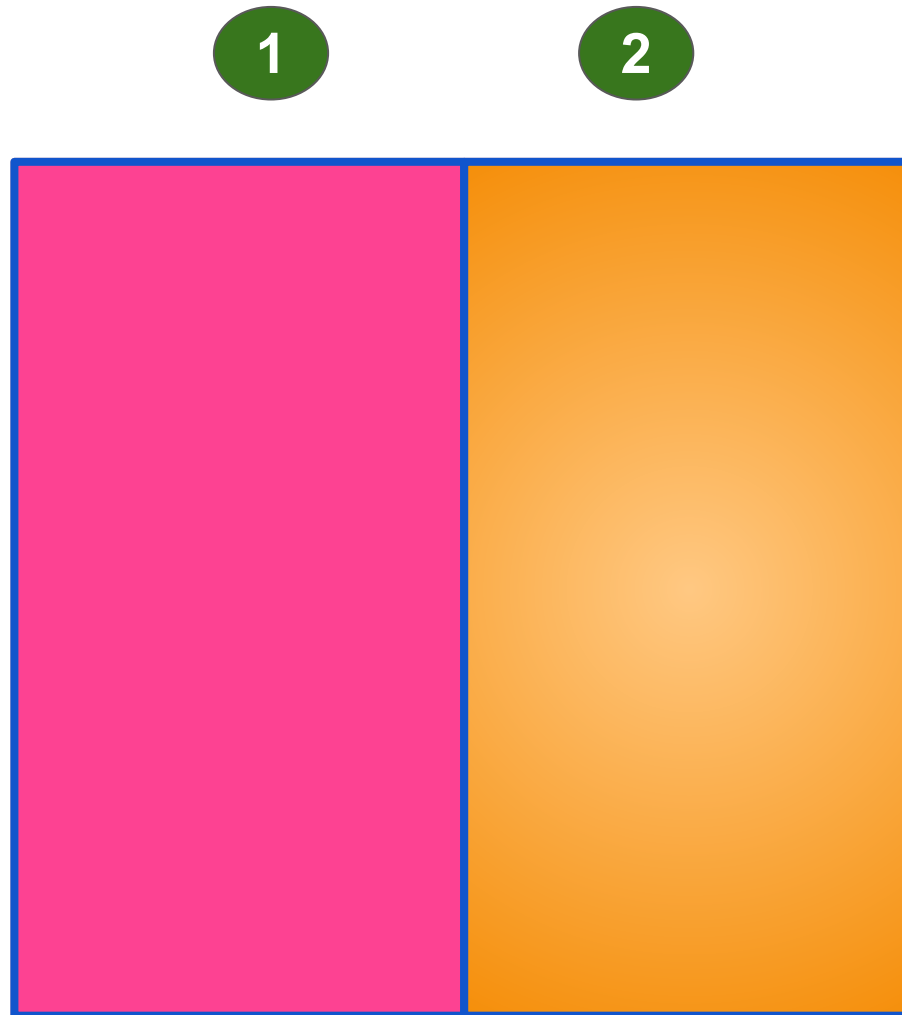


$$\frac{2}{4}$$

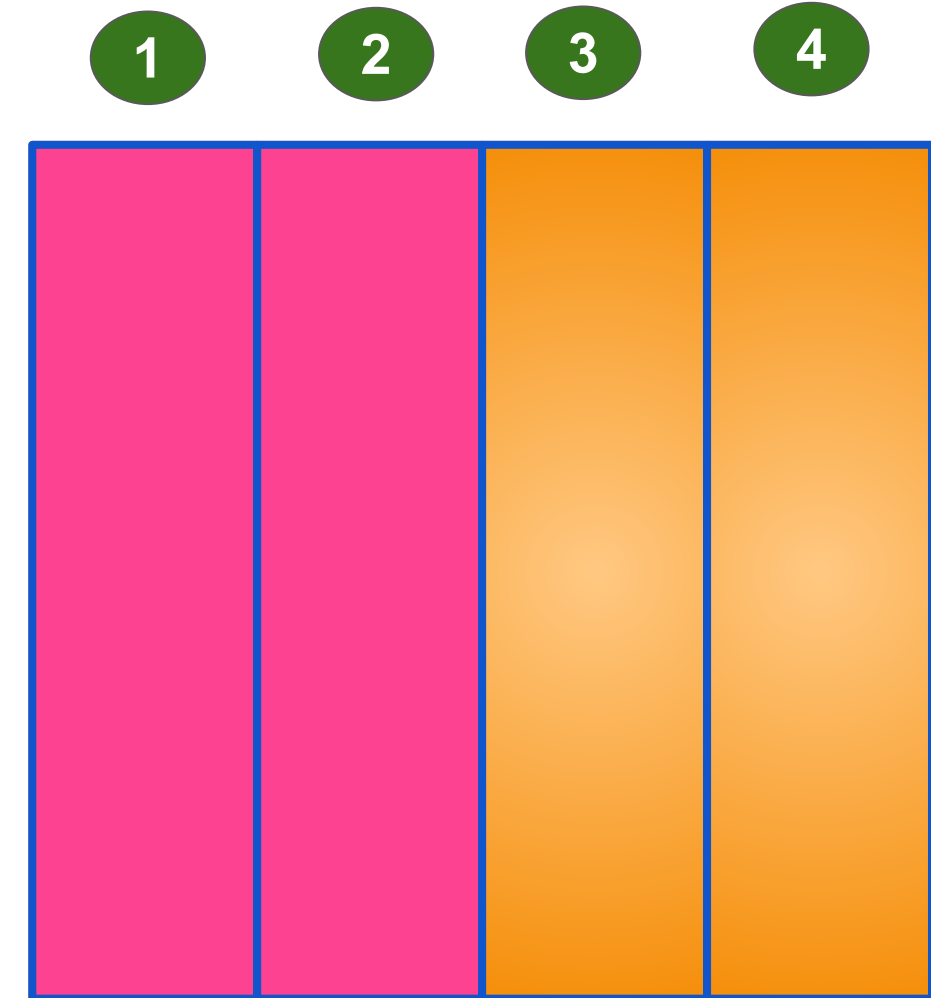
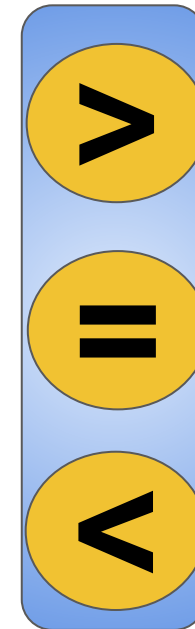
Tap to know.

Next

We do not
compare only the
numbers.
We compare the
shaded amount in
same-sized
wholes.



$$\frac{1}{2}$$

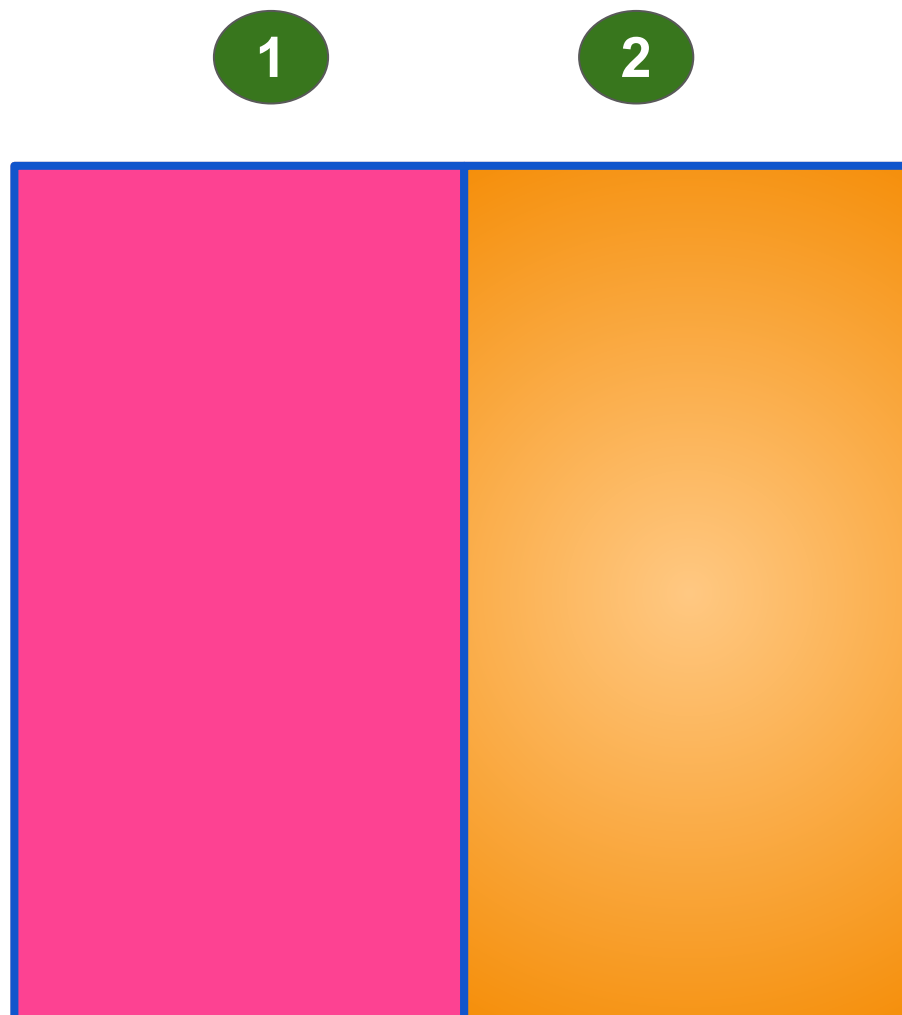


$$\frac{2}{4}$$

Tap the correct symbol.

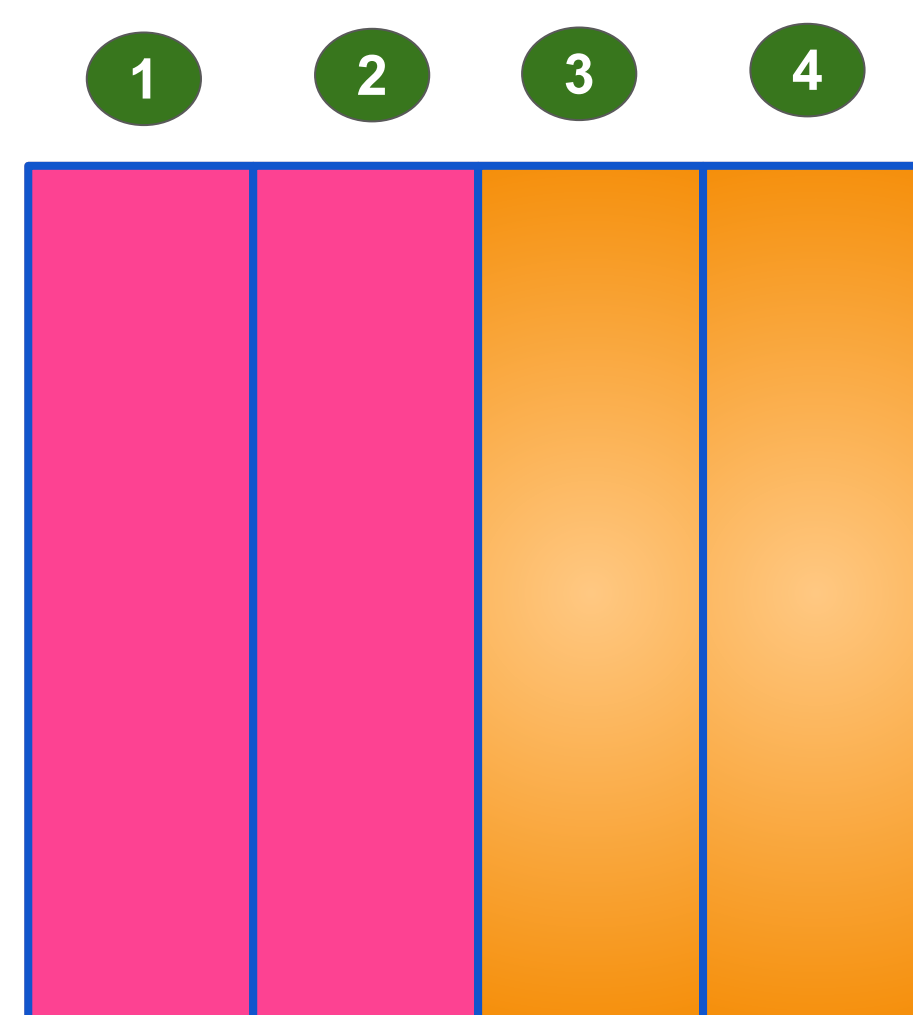
Next

Look at the shaded parts.
Check if they cover the **same amount** of the **whole**?



$$\frac{1}{2}$$

Animate the shaded part from the $\frac{1}{2}$ whole sliding over or overlaying the shaded part in the $\frac{2}{4}$ whole. Use transparency so students can see that the shaded areas match exactly.

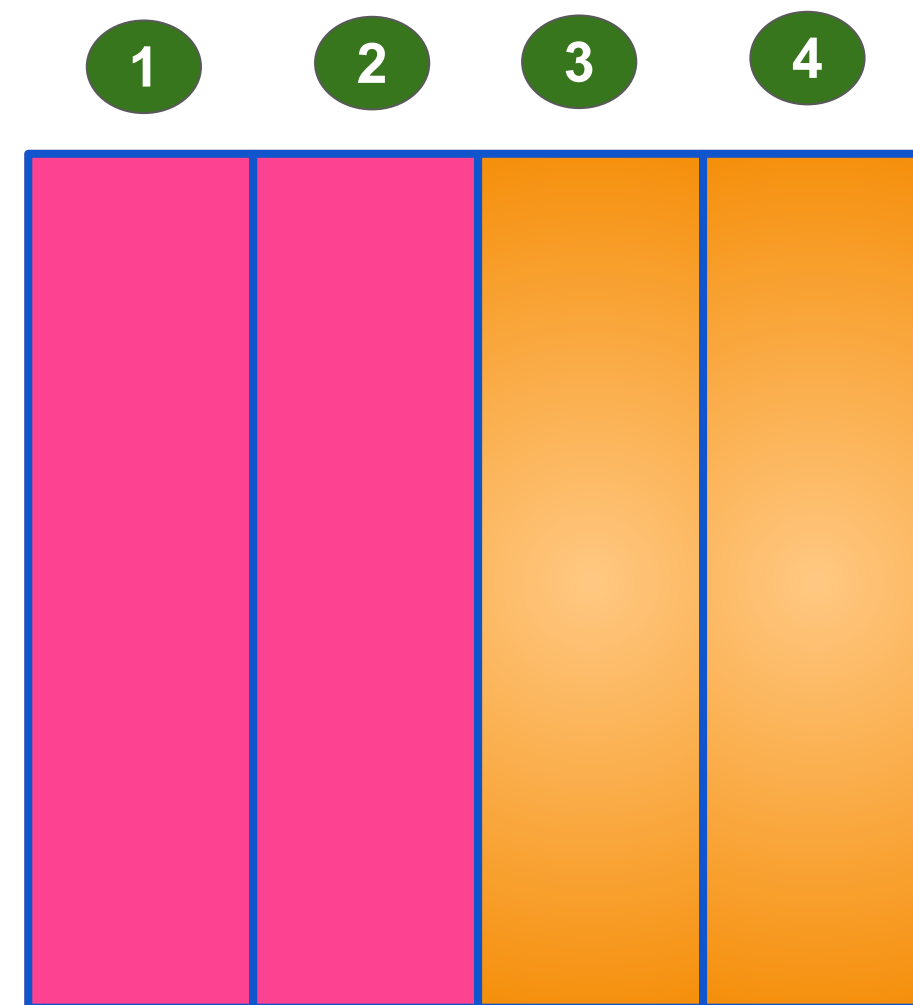
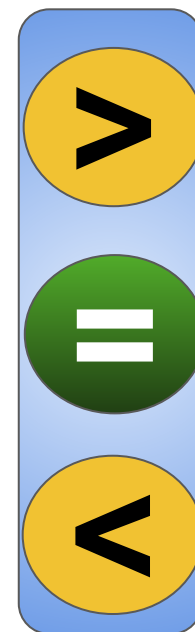
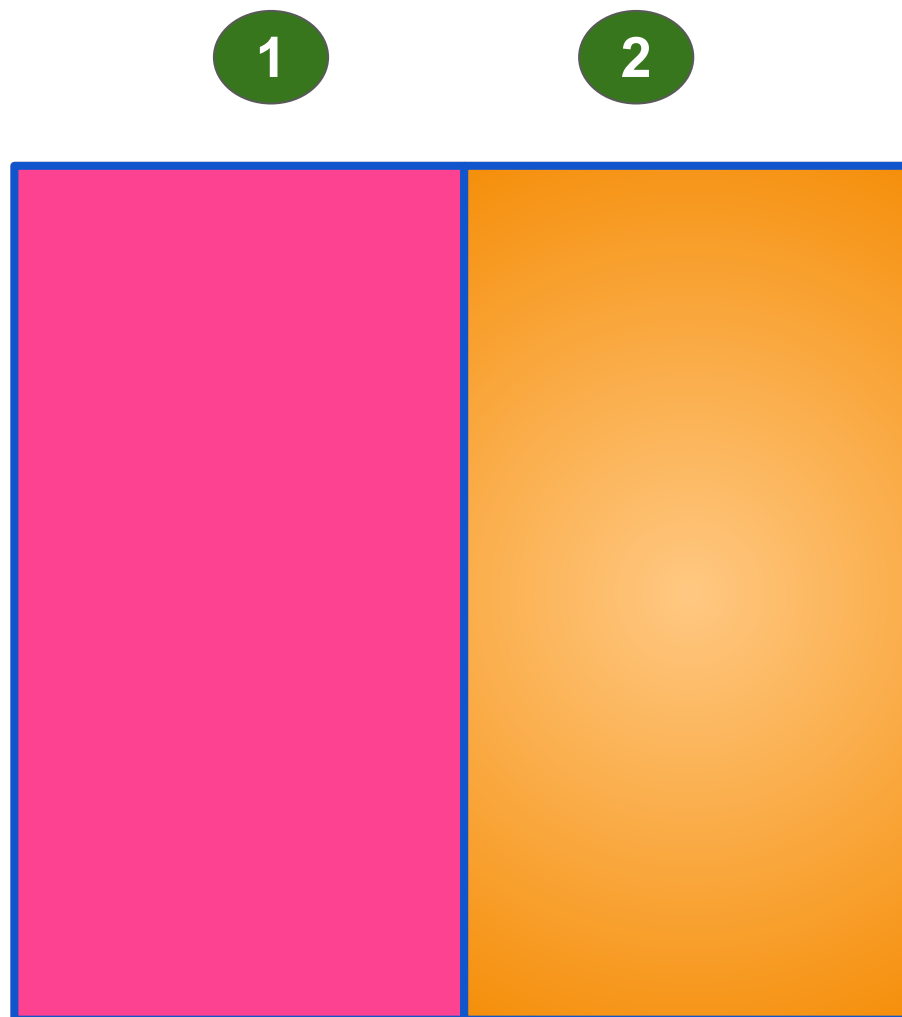


$$\frac{2}{4}$$

Tap the correct answer.

Next

The **shaded parts**
are equal in size.



$$\frac{1}{2}$$

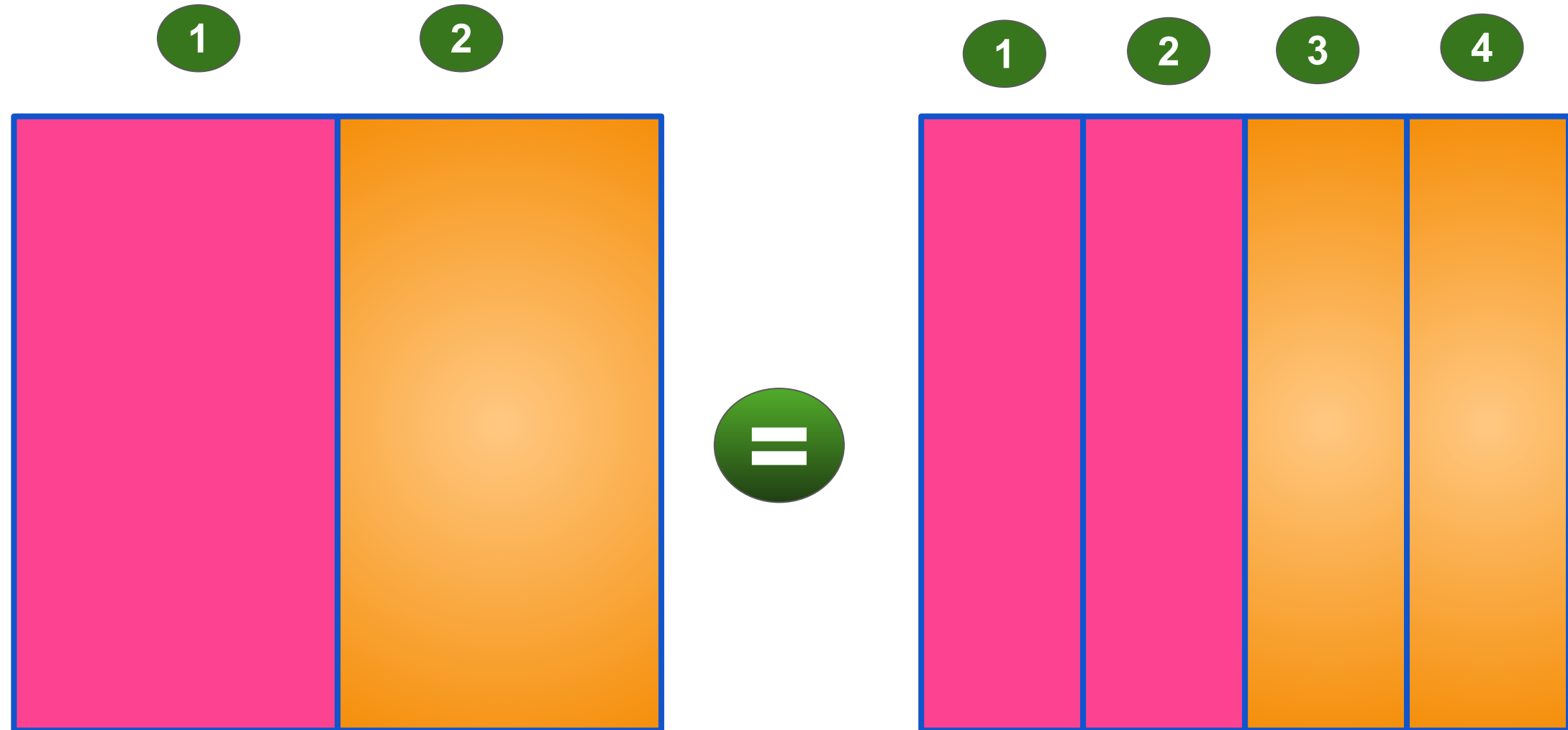
Animate the shaded part from the $\frac{1}{2}$ whole sliding over or overlaying the shaded part in the $\frac{2}{4}$ whole. Use transparency so students can see that the shaded areas match exactly.

$$\frac{2}{4}$$

Tap next to continue.

Next

Even though the **wholes** are divided into **different numbers of equal parts**, the **shaded amount did not change**. Only the number of **equal parts** changed.



$$\frac{1}{2}$$

Pulsate number parts of both the wholes and then pulsate shaded areas.,

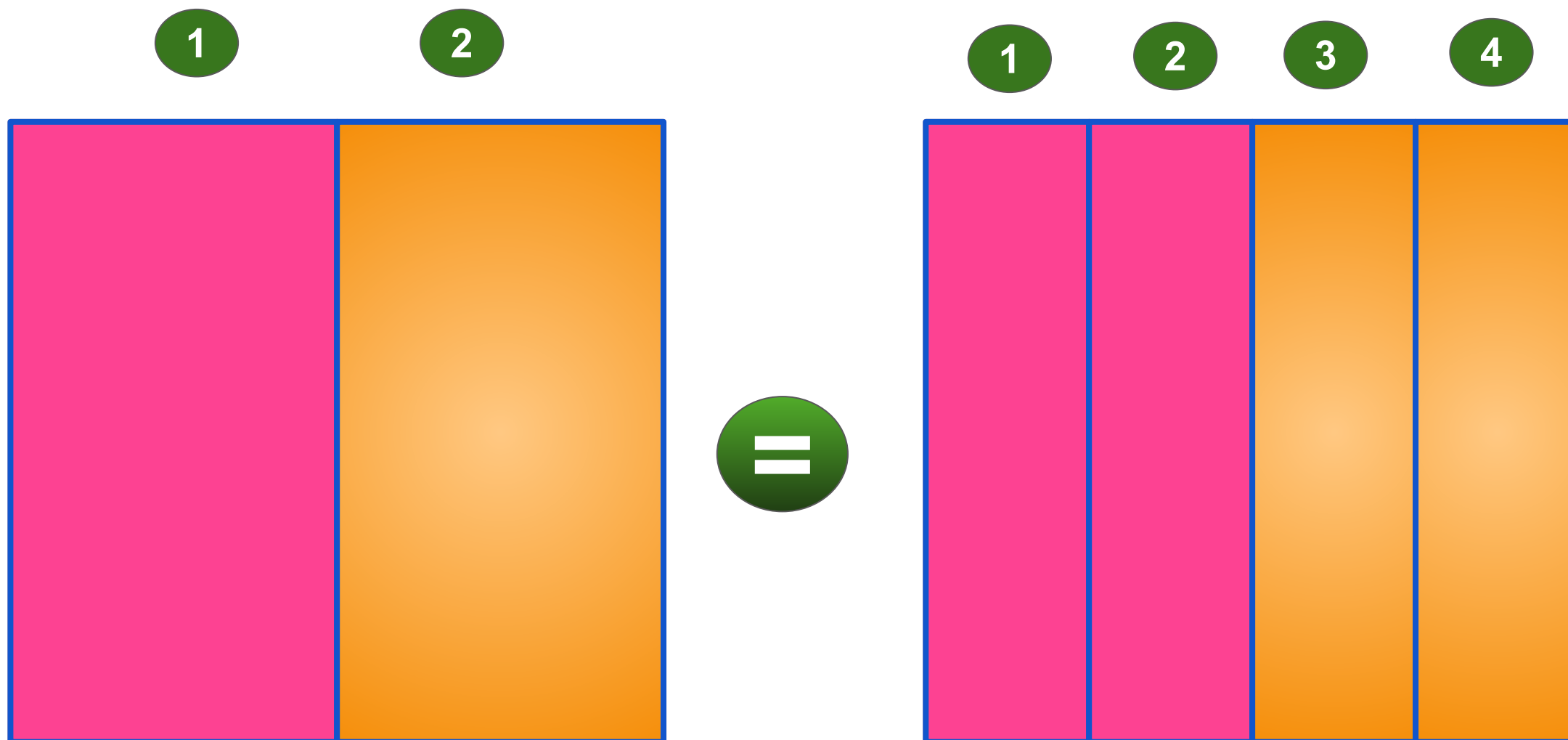


$$\frac{2}{4}$$

Tap next to continue.

Next

So the two different
looking **fractions**
 $\frac{1}{2}$ and $\frac{2}{4}$ represent
the **same amount** of
whole,
so the **fractions** are
equal.



$$\frac{1}{2}$$



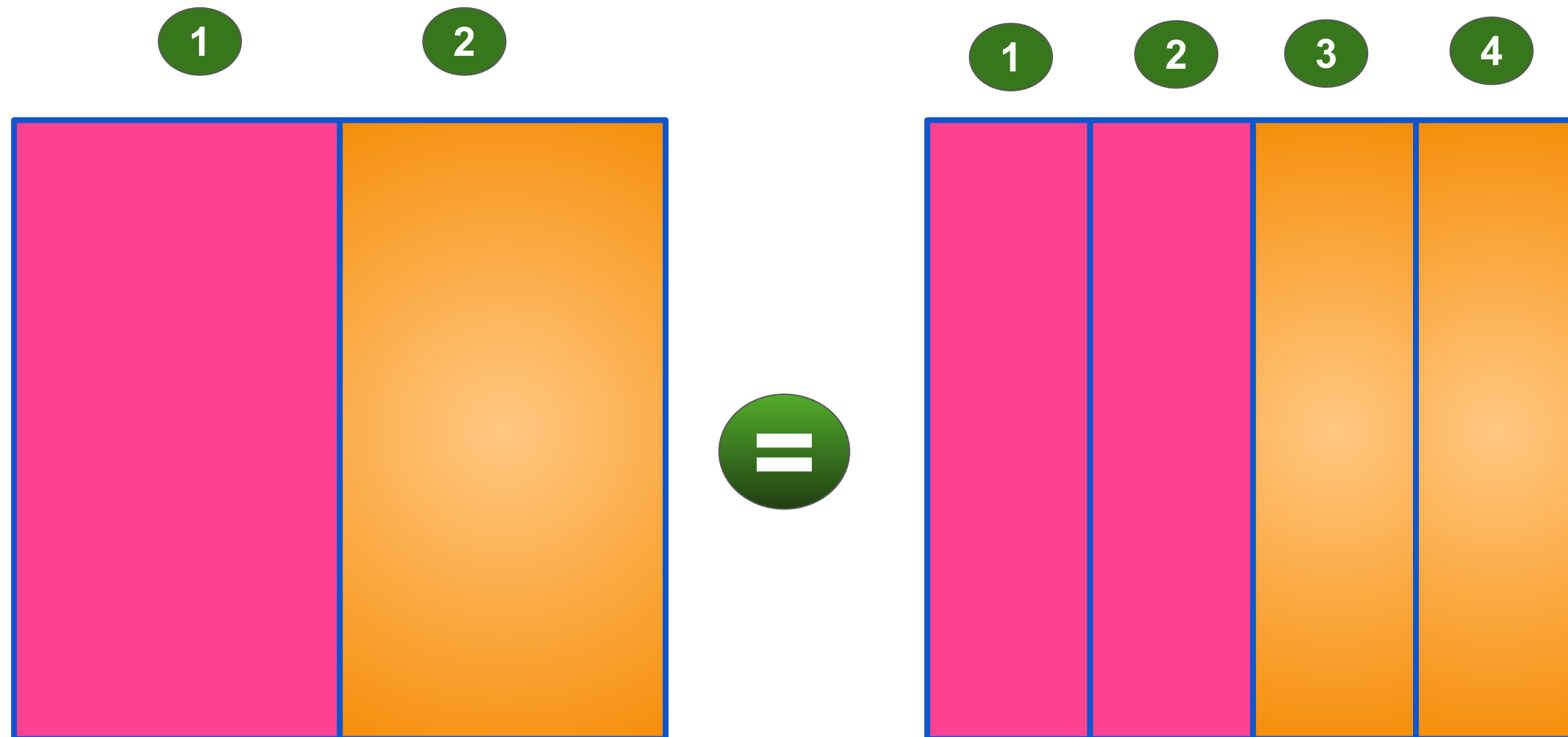
$$\frac{2}{4}$$

Pulsate the shaded areas first,
then pulsate the fraction $\frac{1}{2}$ and
 $\frac{2}{4}$

Tap next to continue.

Next

The **fractions**
 $\frac{1}{2}$ and $\frac{2}{4}$ are
called
equivalent
fractions.



Pulsate $\frac{1}{2}$ and $\frac{2}{4}$

$$\frac{1}{2}$$

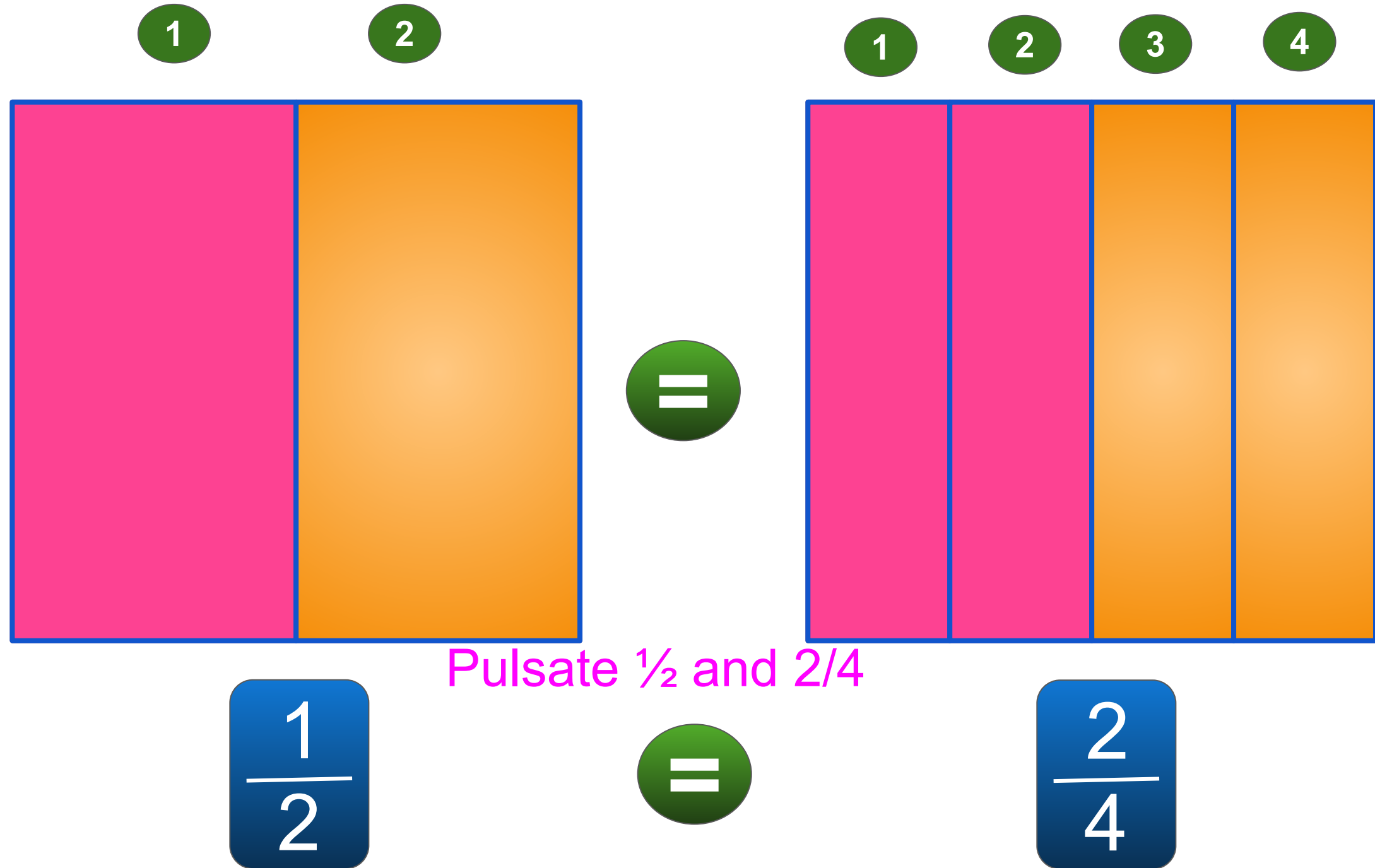
=

$$\frac{2}{4}$$

Tap Next to continue.

Next

Fractions that look different but show the same amount are called **equivalent fractions**.



Tap Next to explore another pair of equivalent fractions.

Next



Follow the same loop as followed
for 4 equal parts if the user
choose 6 equal parts.

Equivalent Fractions

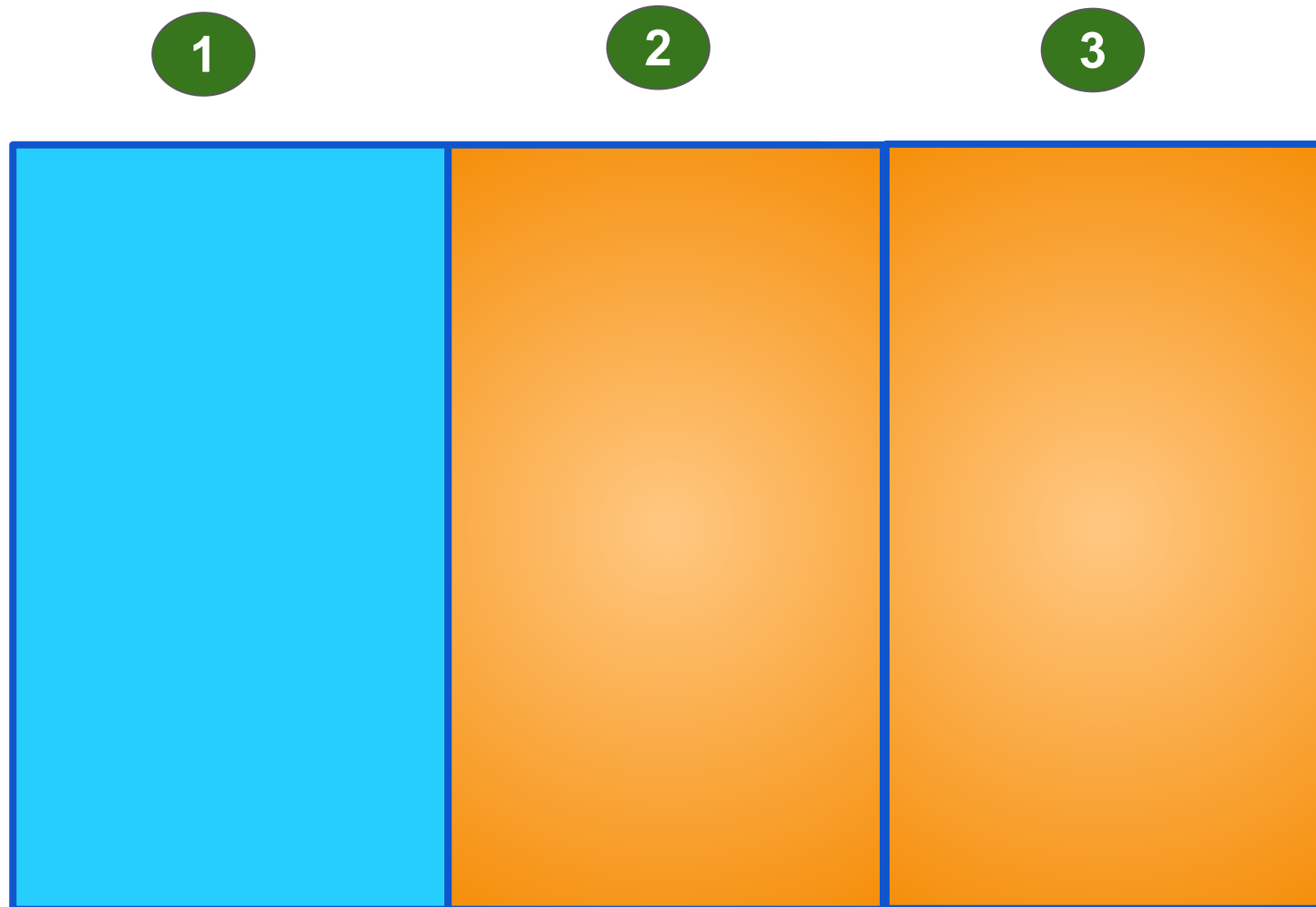


Are $\frac{1}{3}$ and $\frac{2}{6}$ **equivalent fractions** ?

Let's find out!

Next

This whole
shows the
fraction $\frac{1}{3}$.

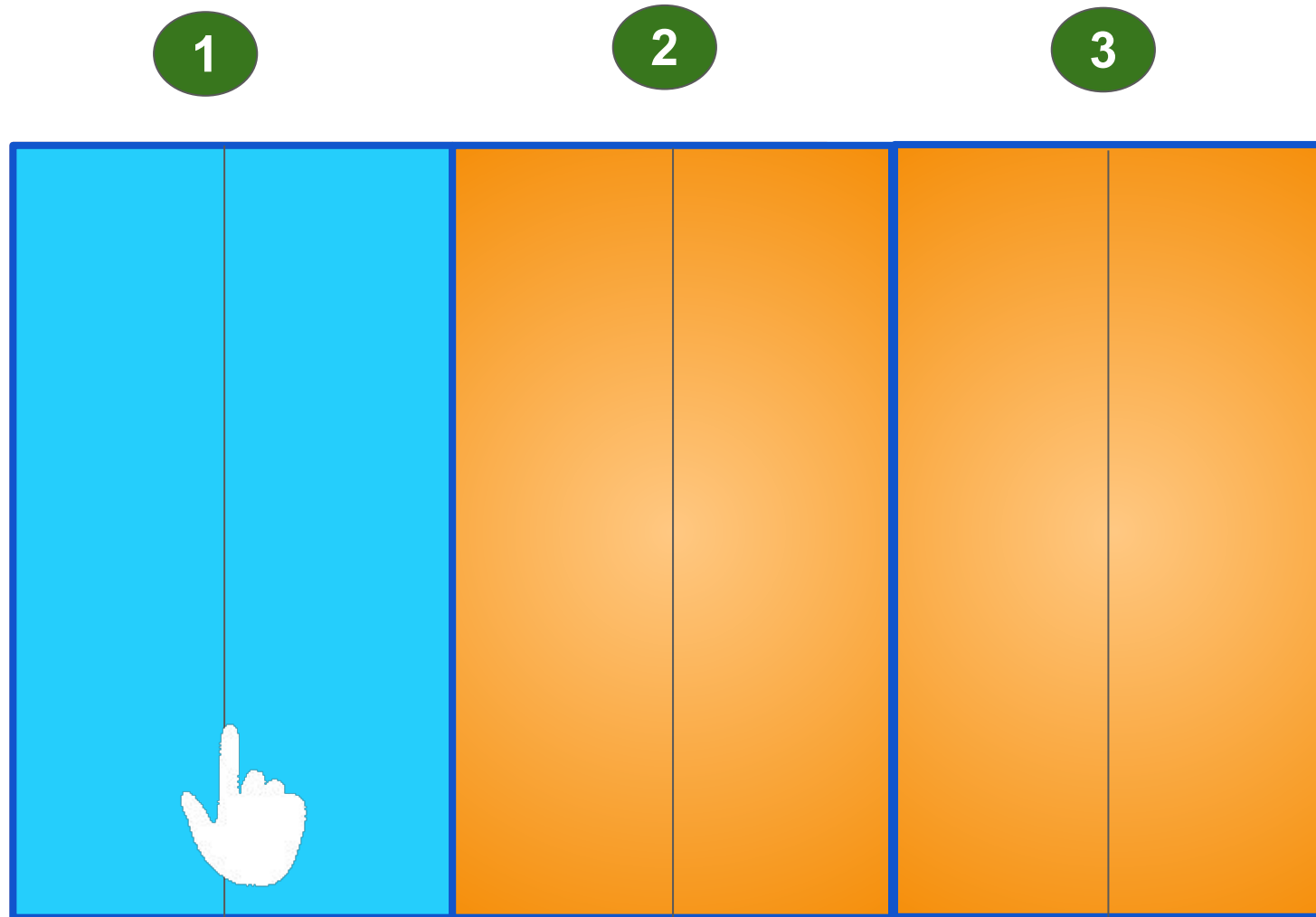


$$\frac{1}{3}$$

Tap Next to divide the whole further.

Next

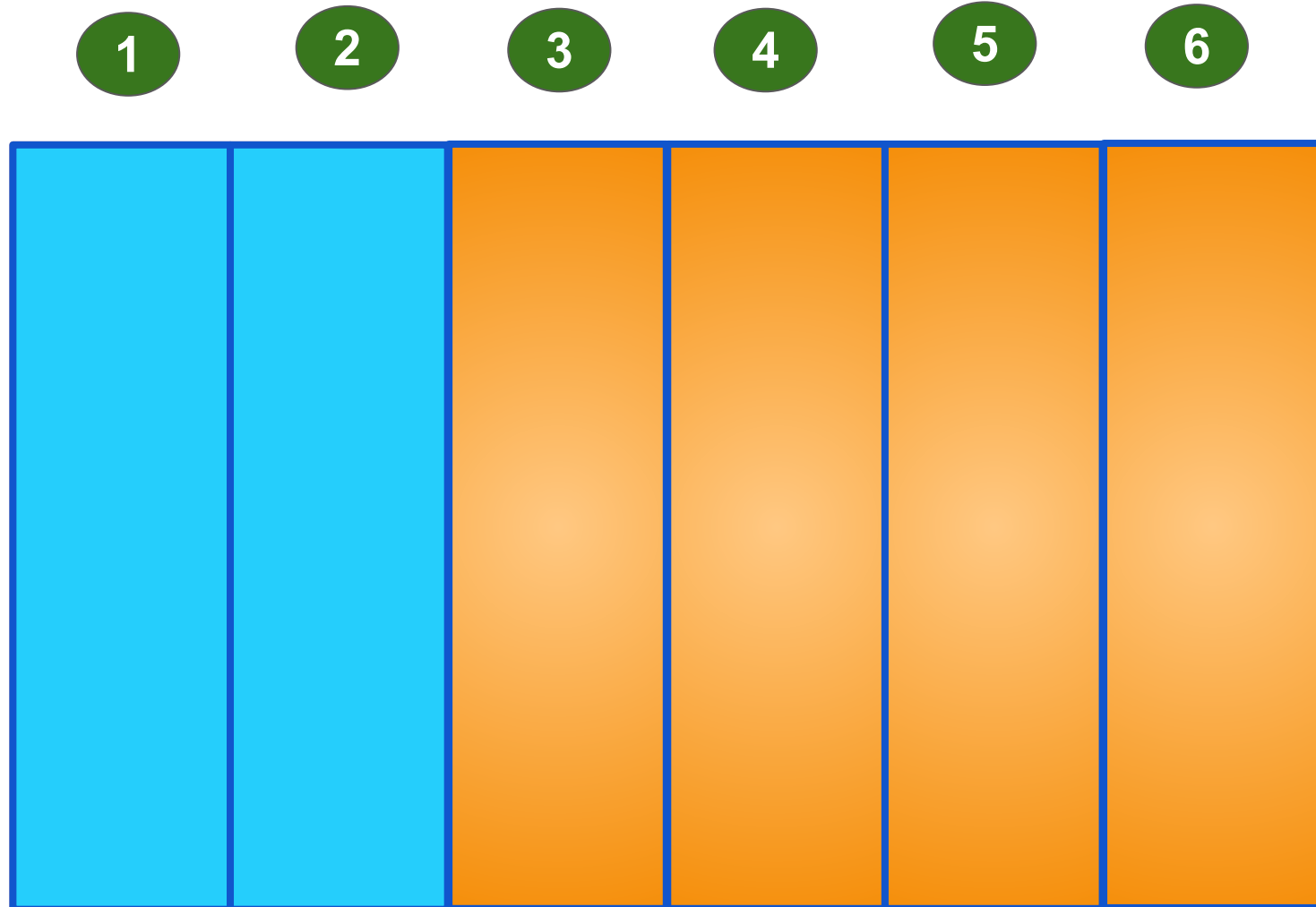
Let us divide
each **third** into
2 smaller equal
parts.



$$\frac{1}{3}$$

Tap the dotted lines to split each third.

What fraction is shaded now?



$$\frac{1}{3}$$

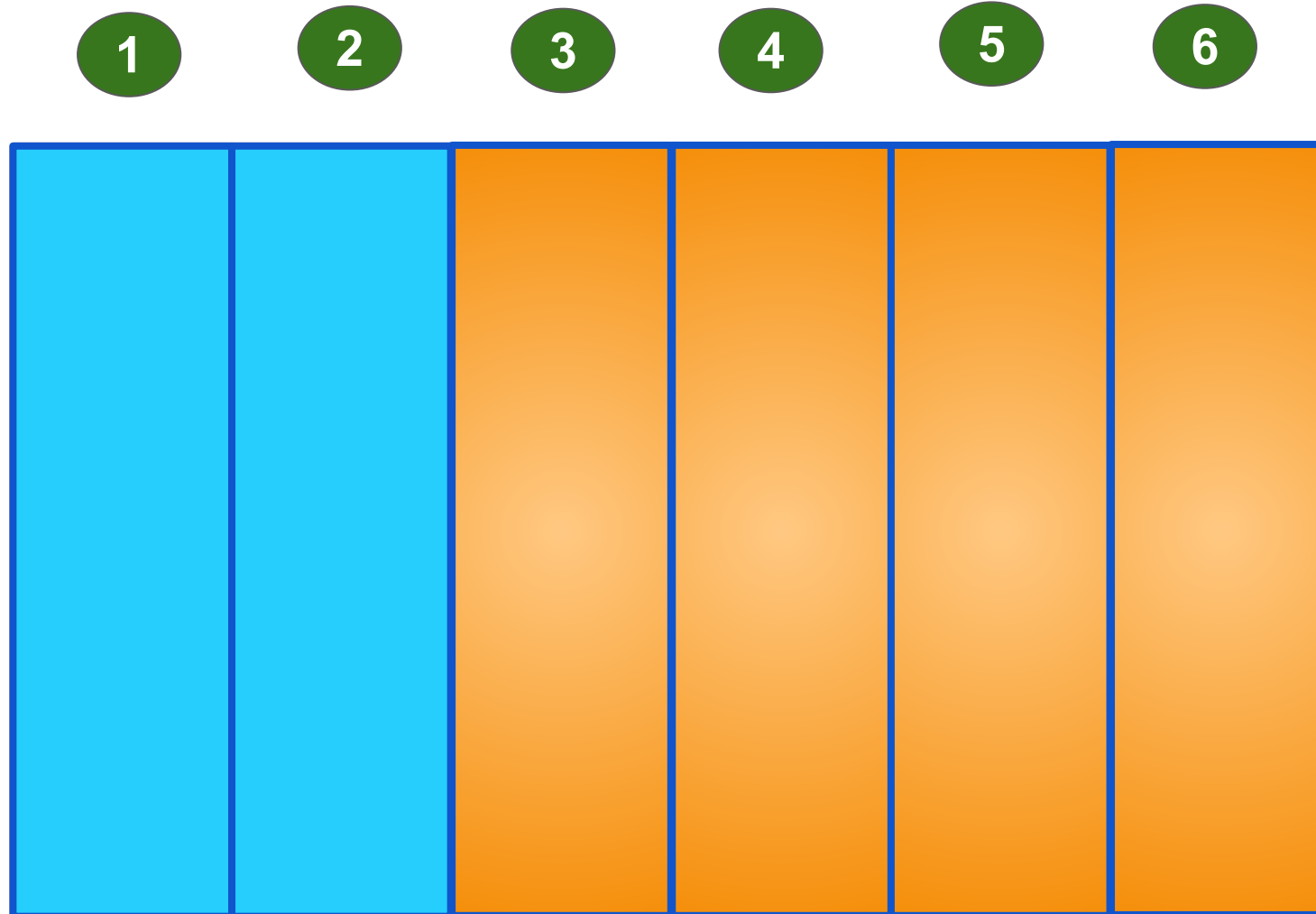
$$\frac{1}{6}$$

$$\frac{3}{6}$$

$$\frac{2}{6}$$

Tap the correct answer.

Let's count again.
Check how many
parts are shaded?
And how many
equal parts are
there in all?



$$\frac{1}{3}$$

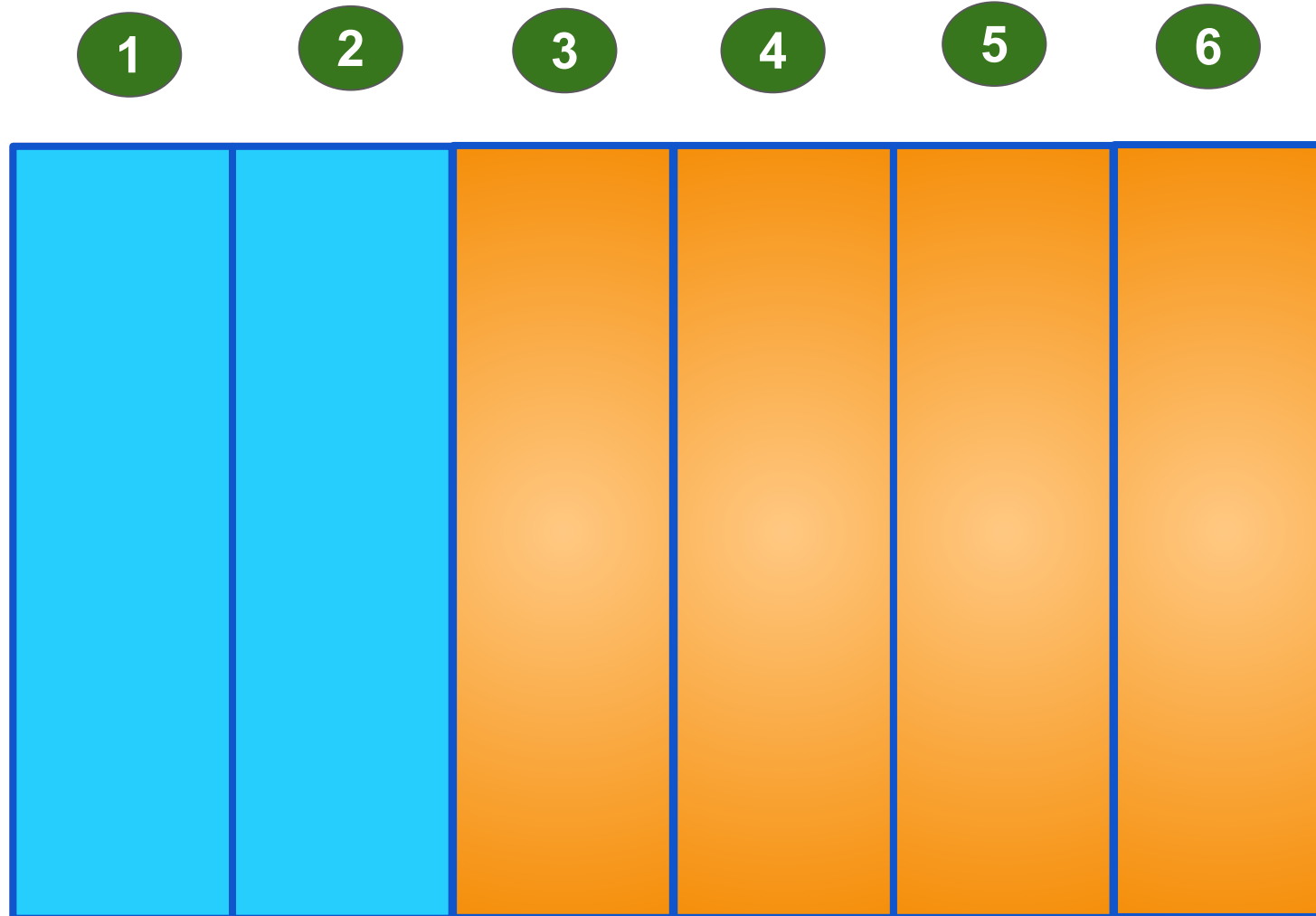
$$\frac{1}{6}$$

$$\frac{3}{6}$$

$$\frac{2}{6}$$

Tap the correct answer.

Yes! 2 out of 6
equal parts are
shaded



$$\frac{1}{3}$$

$$\frac{2}{6}$$

$$\frac{1}{6}$$

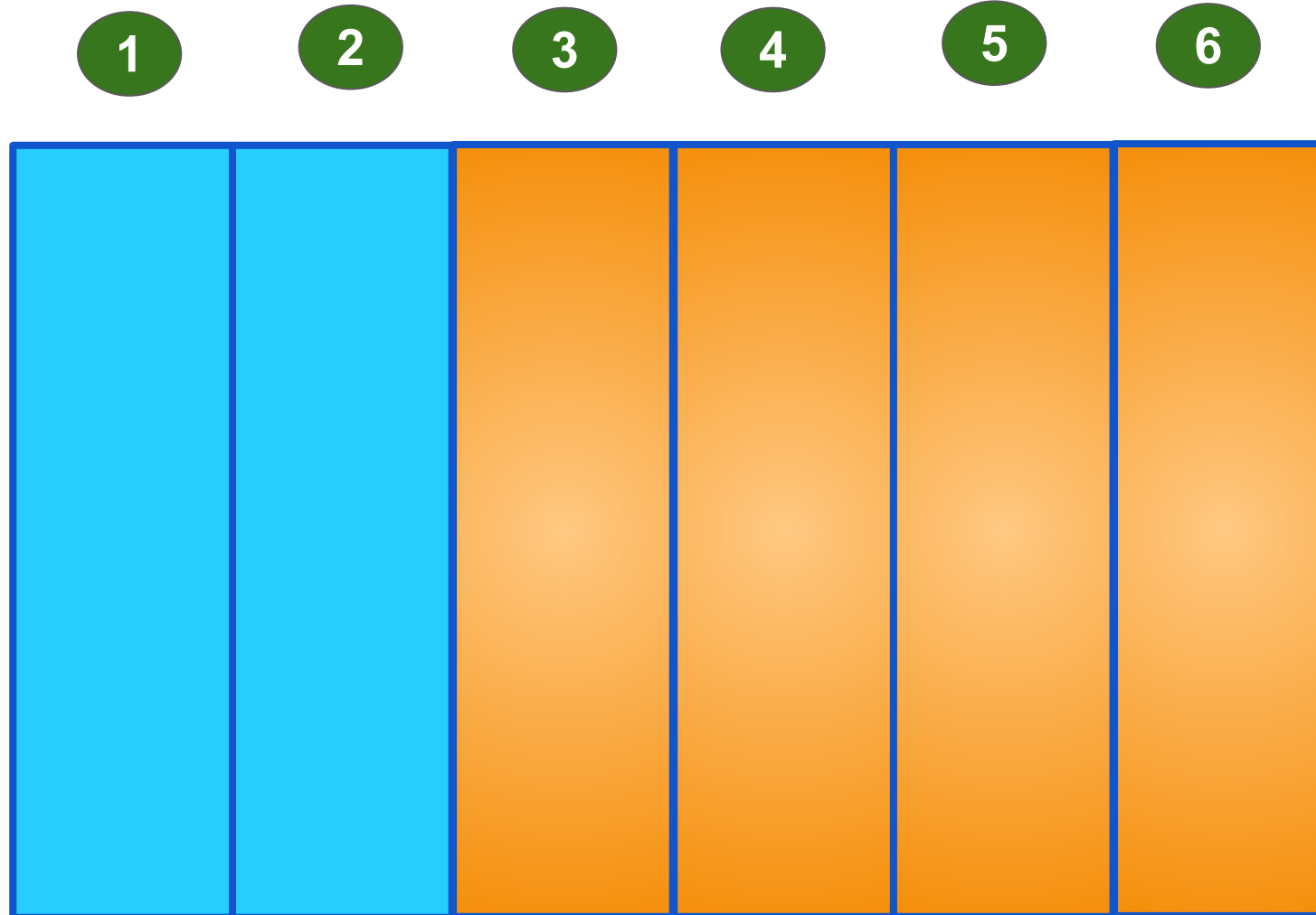
$$\frac{3}{6}$$

$$\frac{2}{6}$$

Tap Next to continue.

Next

Do $\frac{1}{3}$ and $\frac{2}{6}$
show the
same amount?



$$\frac{1}{3}$$

?

$$\frac{2}{6}$$

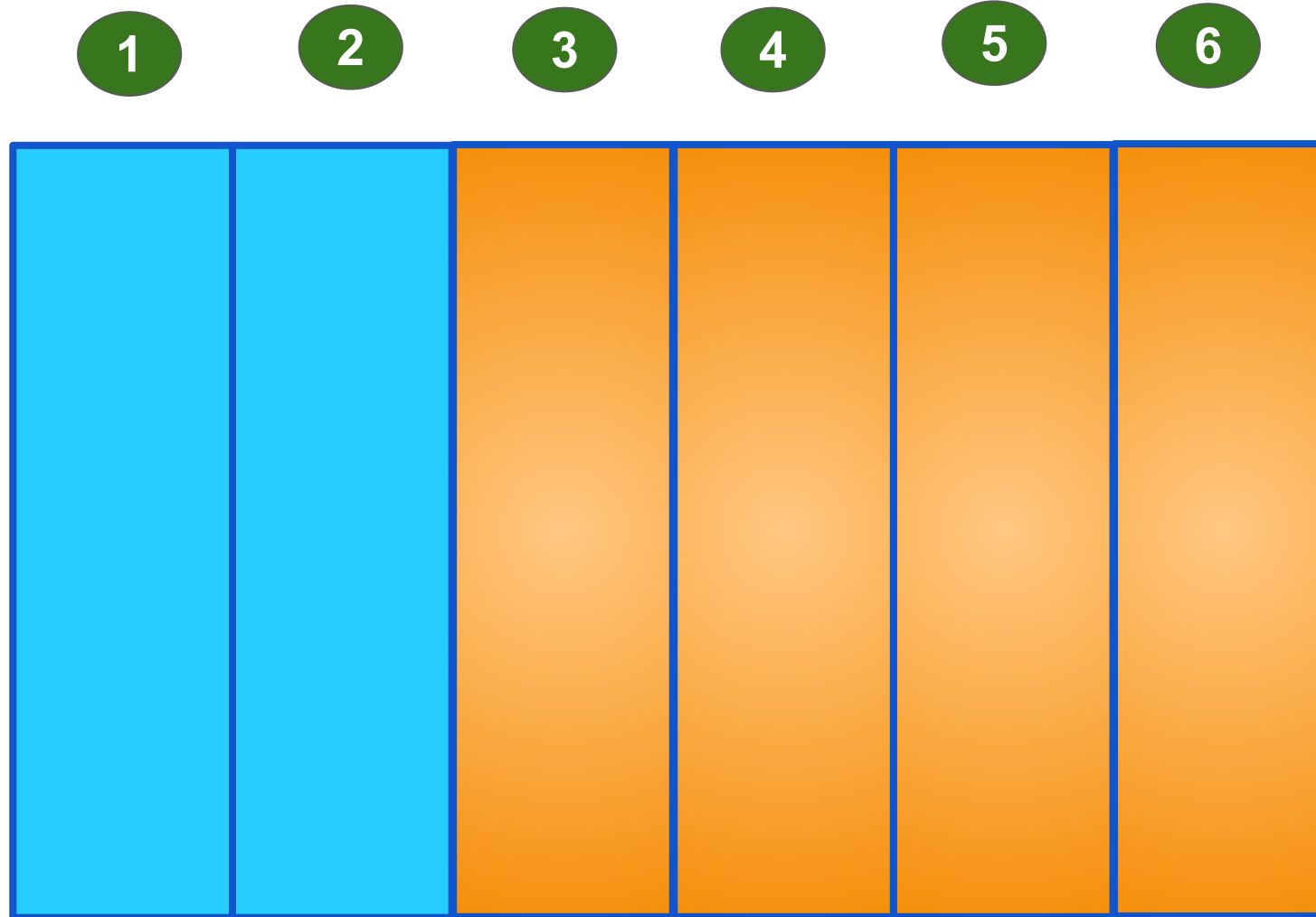
Yes

No

Tap the correct answer

Look at the **shaded parts** again.

Even though the **whole** is divided into smaller equal parts, the **shaded amount** didn't change.



$$\frac{1}{3}$$

?

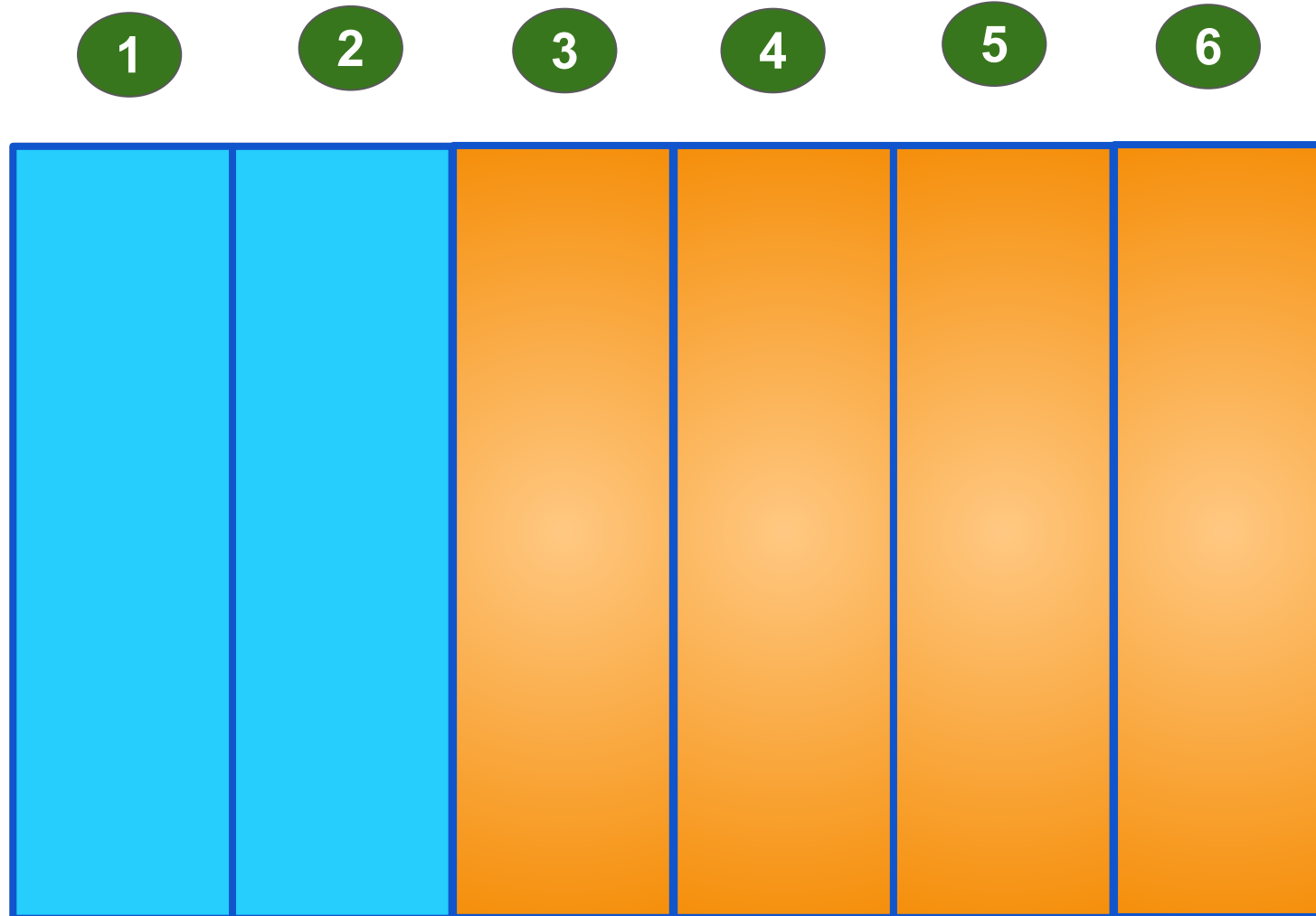
$$\frac{2}{6}$$

Yes

No

Tap the correct answer

The parts became smaller, but the shaded amount did not change.
So the fractions show the same amount.



$$\frac{1}{3} = \frac{2}{6}$$

Once the correct option is chosen, replace the question mark by “=” and then enable the next button

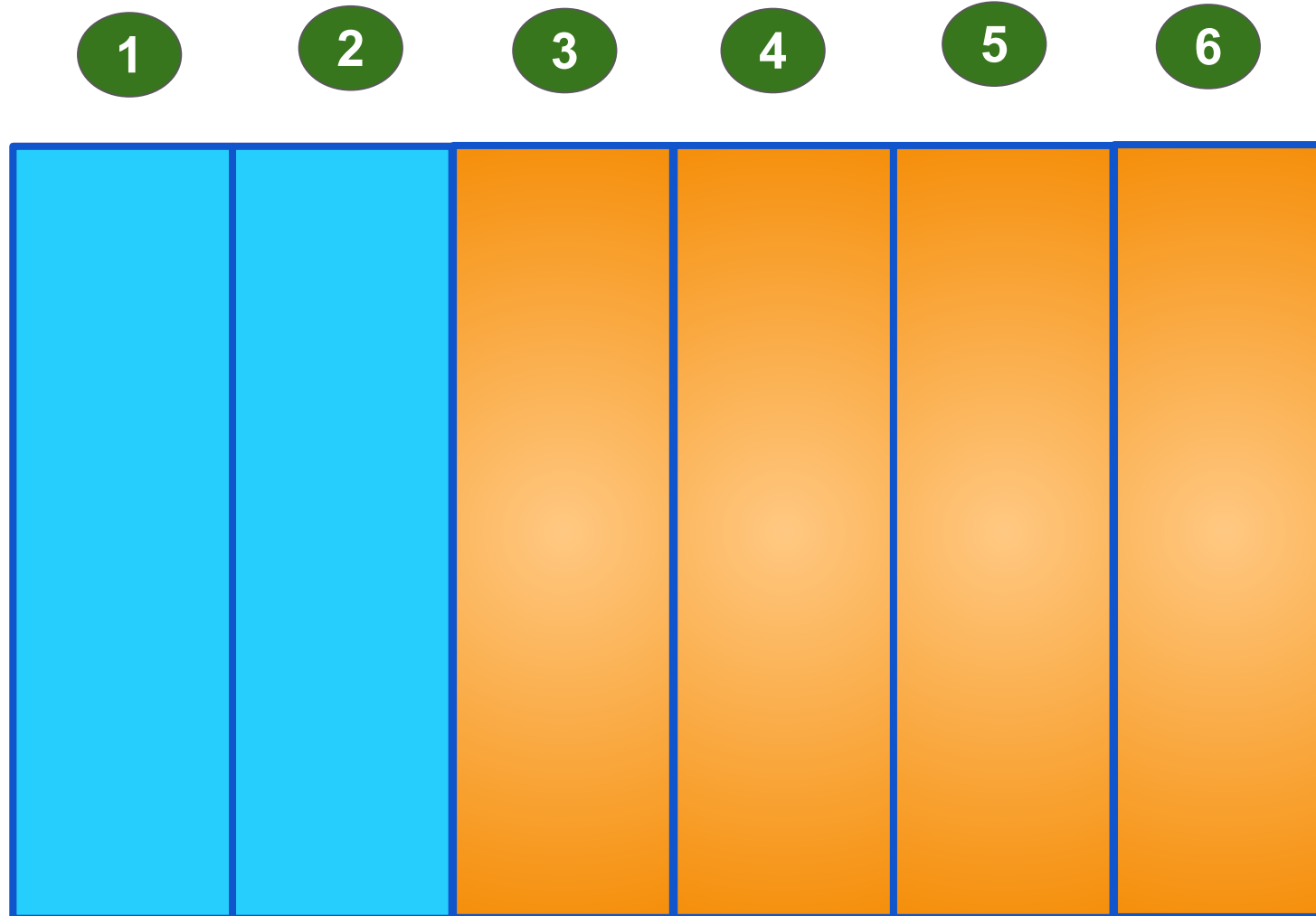
Yes

No

Tap Next to continue.

Next

Fractions
 $\frac{1}{3}$ and $\frac{2}{6}$ are
also **equivalent**
fractions.



$$\frac{1}{3}$$

$$=$$

$$\frac{2}{6}$$

Equivalent fractions may
look different, but they
represent the same amount.

Tap Next to continue.

Next

Equivalent Fractions



Fractions that look different but show the **same amount of the whole** are called **equivalent fractions**.

$$\frac{1}{2} = \frac{2}{4}$$

$$\frac{1}{3} = \frac{2}{6}$$

Equivalent Fractions



Now, can we generate
equivalent fractions ?

Next

Comparing Fractions Visually



That's an interesting
question!
You'll explore this in the next
activity.

Want to try again?

Start Again